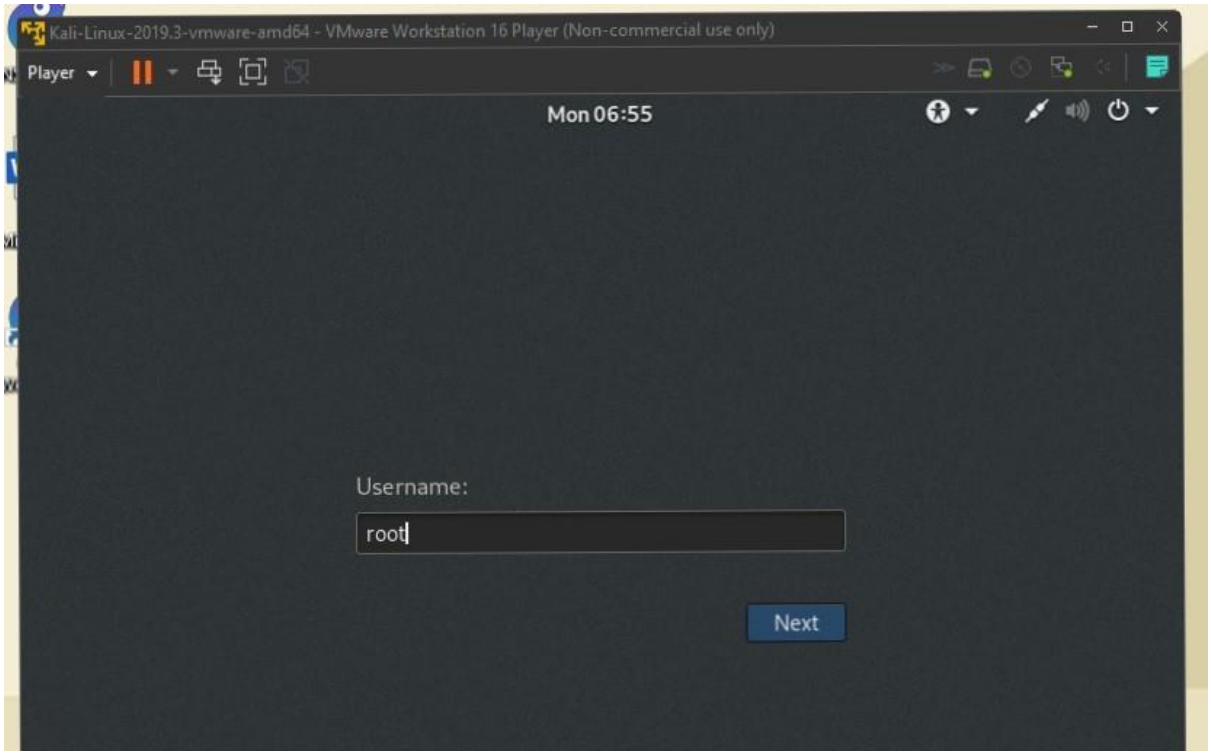
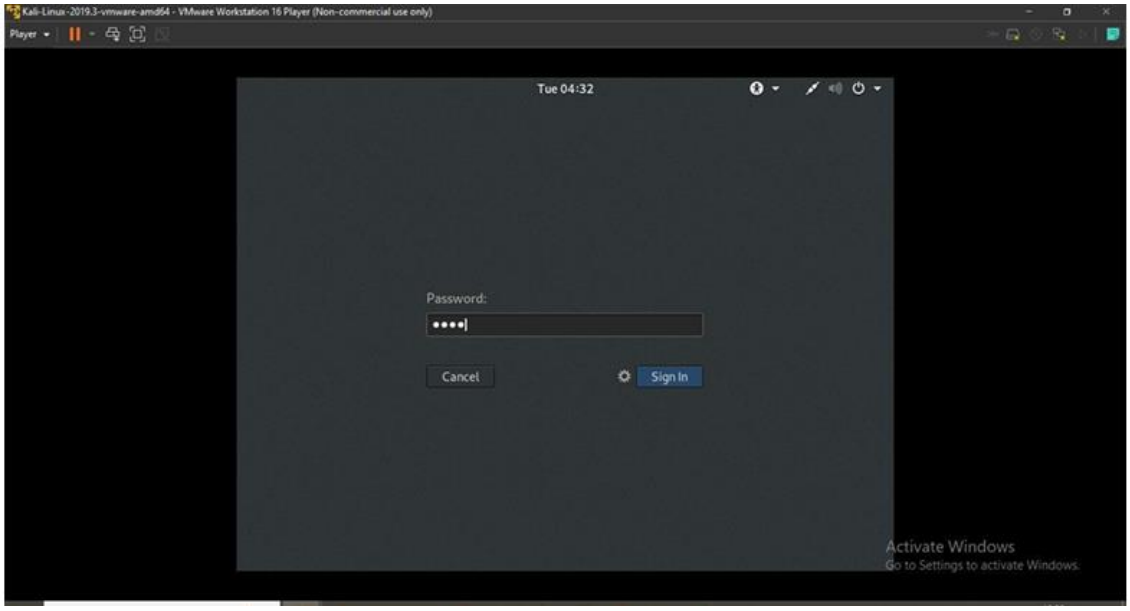
Oratilwe Masango

CySA Project 1

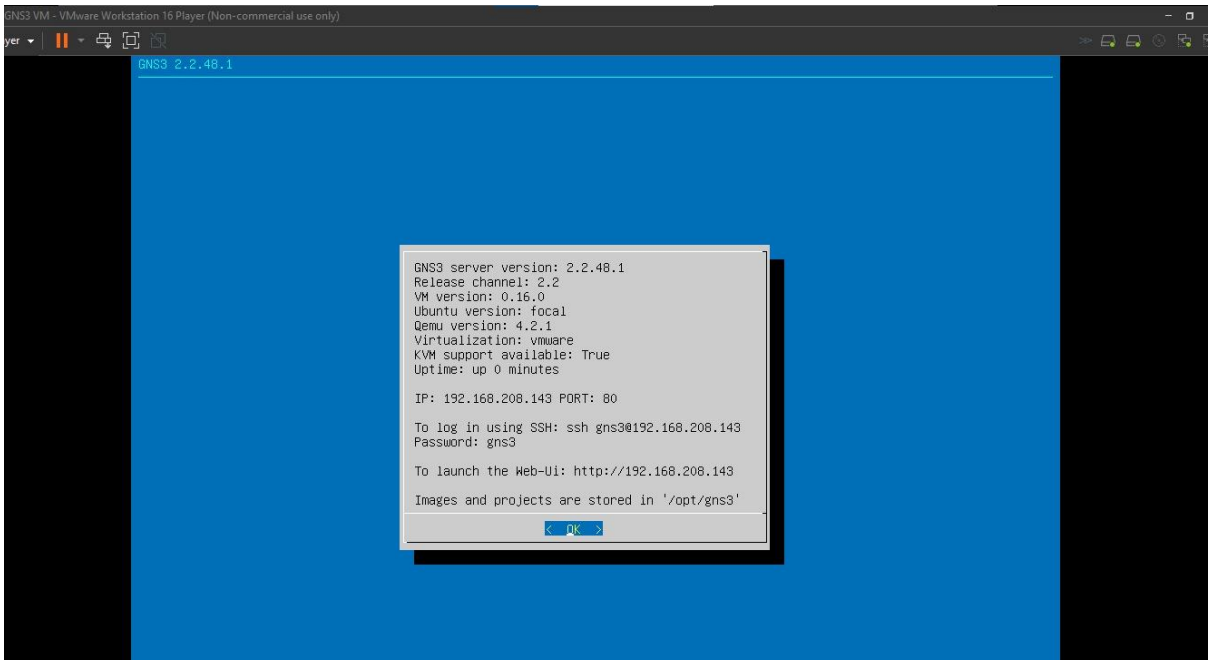
CON-1488359-X2F0

KALI LINUX INSTALLATION

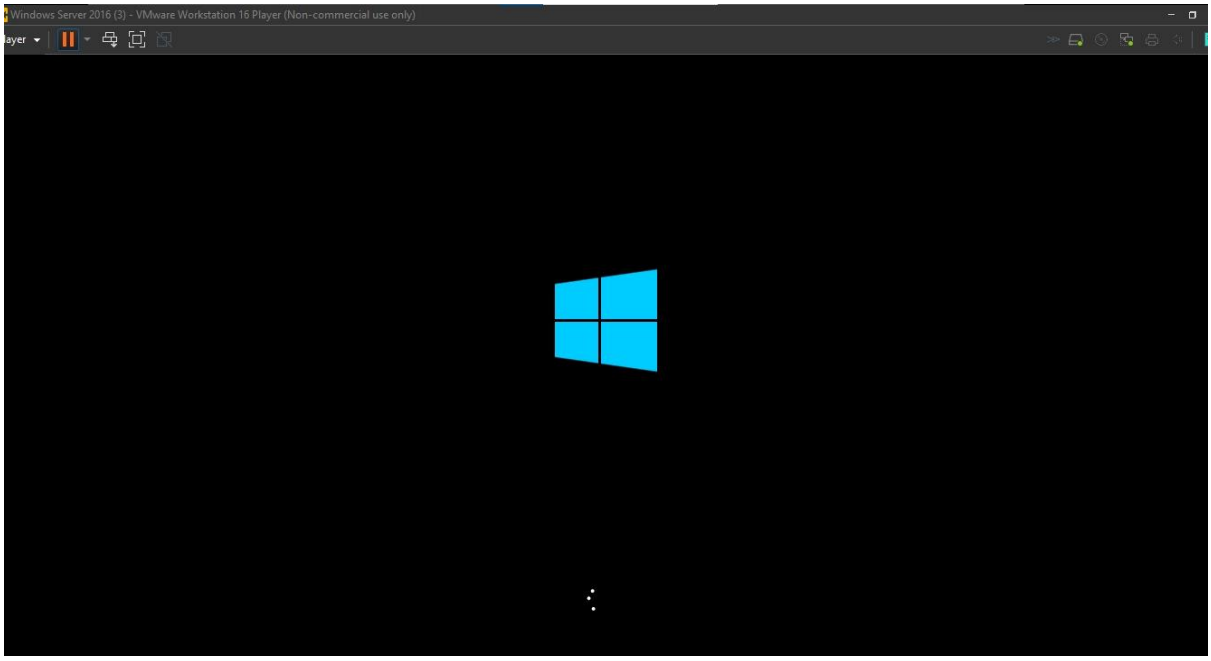


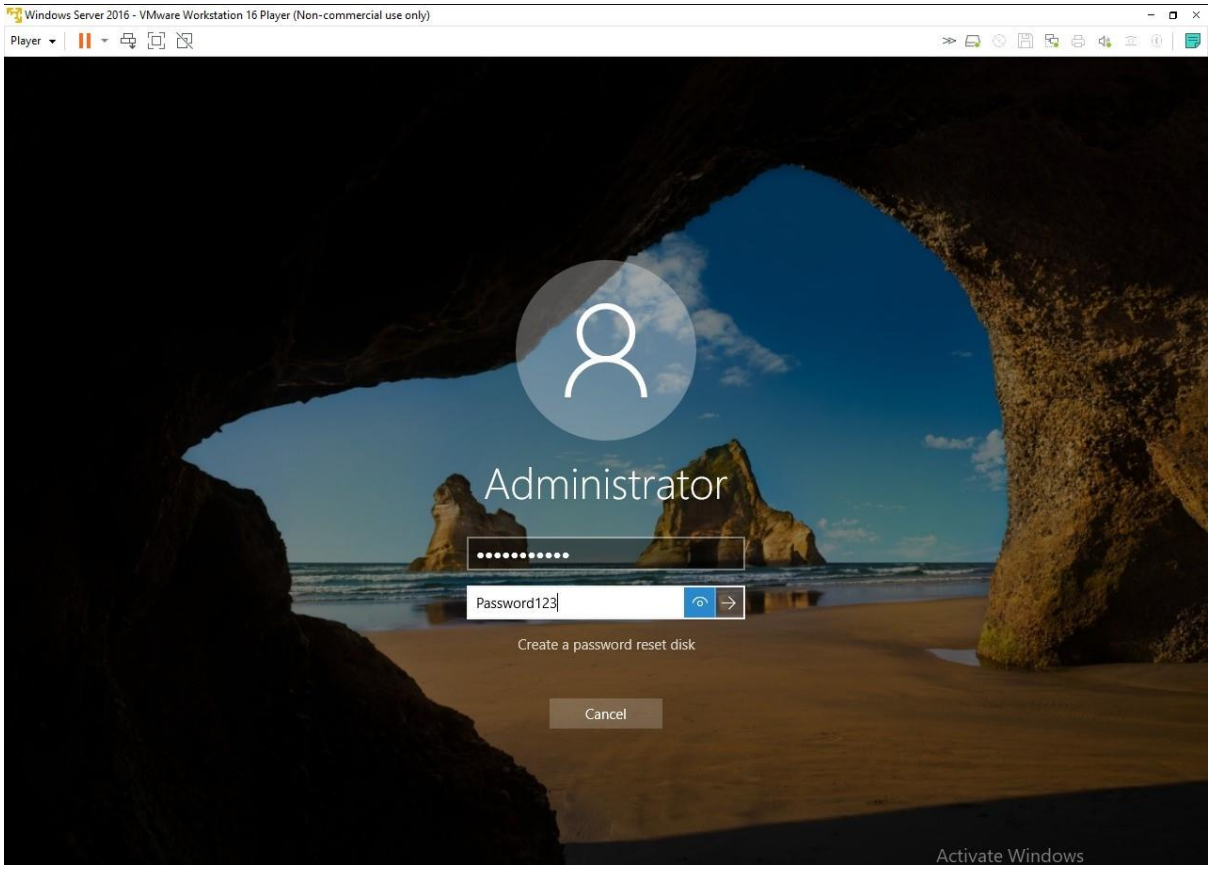


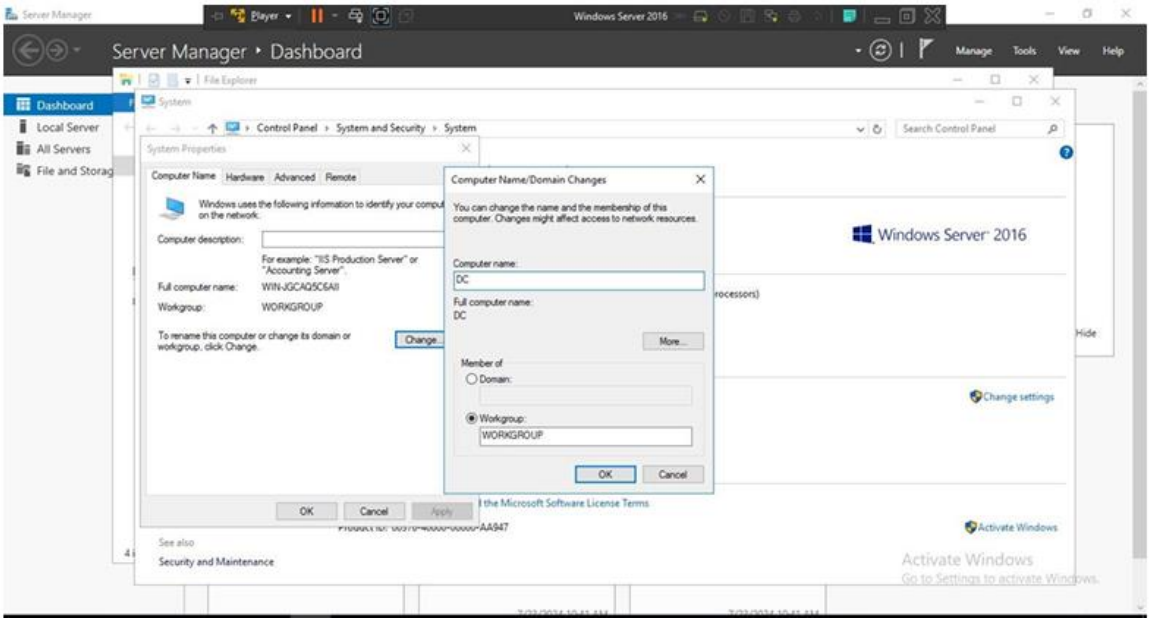
GNS3 INSTALLATION ON VMWARE

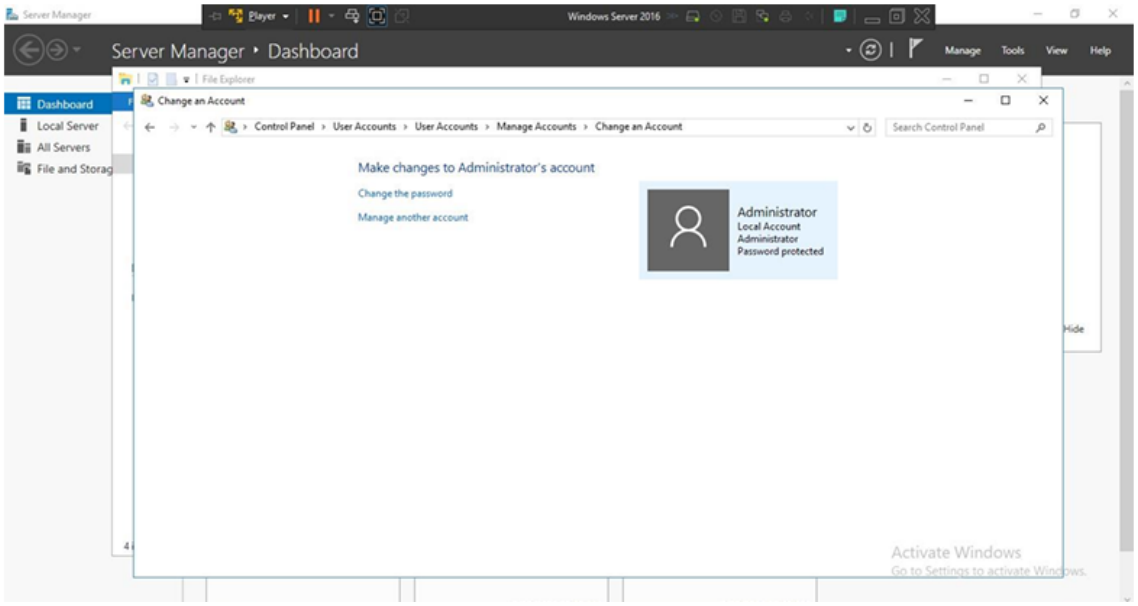


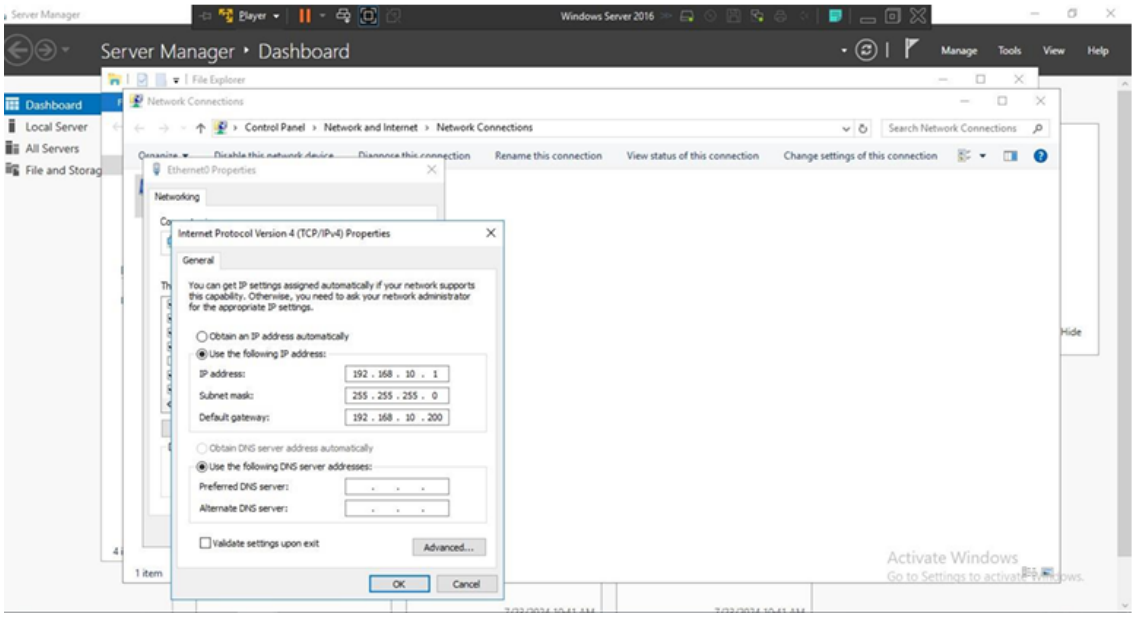
WINDOWS SERVER INSTALLATION

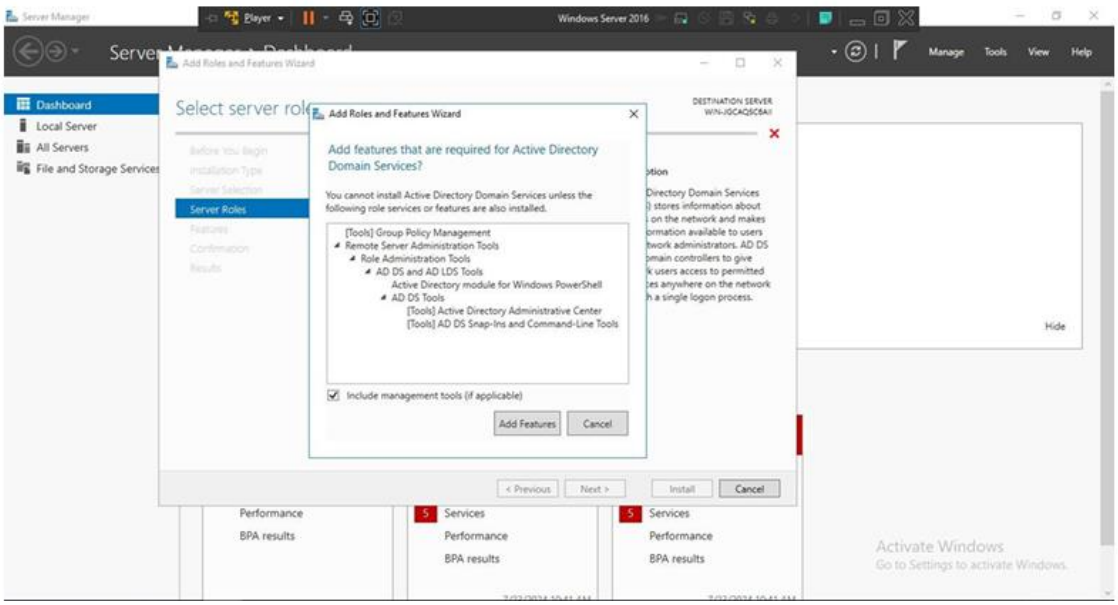


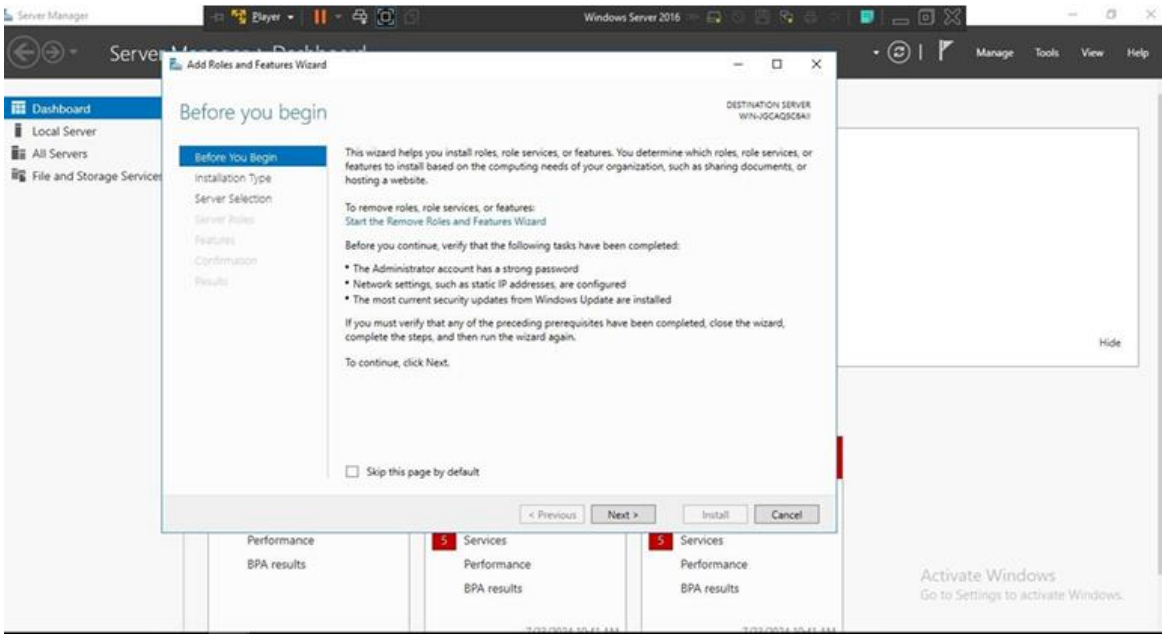

Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Player
Administrator
Password123
Create a password reset disk
Cancel
Activate Windows

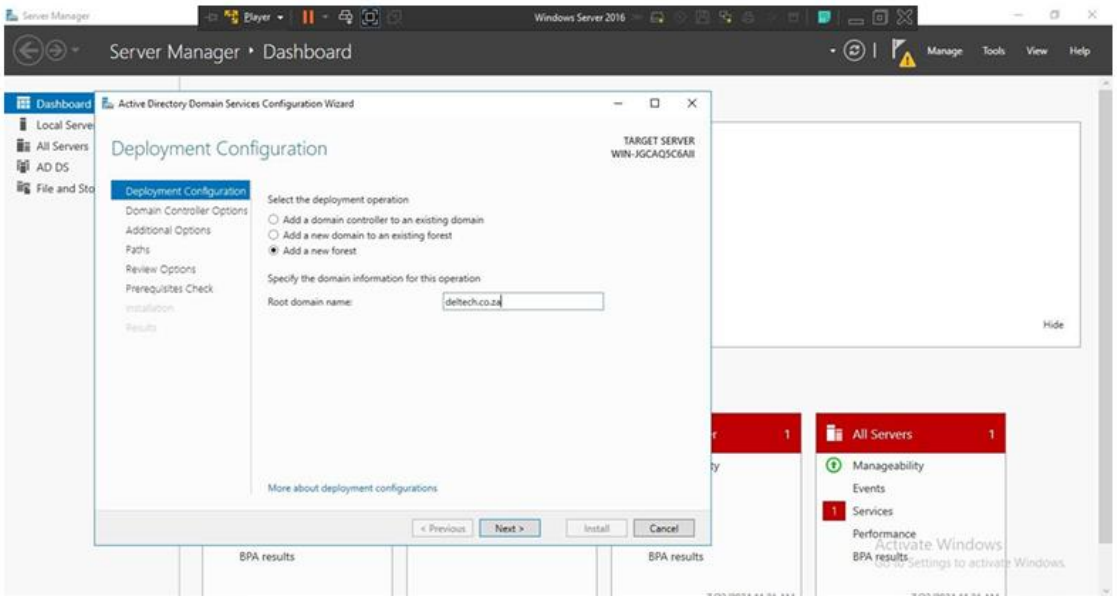

Server Manager
Dashboard
Local Server
All Servers
File and Storag
Manage
Tools
View
Help
Control Panel
System and Security
System
Search Control Panel
System Properties
Computer Name
Hardware
Advanced
Remote
Computer description:
For example: "IIS Production Server" or "Accounting Server".
Full computer name:
WIN-JGCAQ5C6AII
Workgroup:
WORKGROUP
To rename this computer or change its domain or workgroup, click Change.
Change...
Computer Name/Domain Changes
You can change the name and the membership of this computer. Changes might affect access to network resources.
Computer name:
DC
Full computer name:
DC
More...
Member of
Domain:
Workgroup:
WORKGROUP
OK
Cancel
Windows Server 2016
Change settings
Activate Windows
Security and Maintenance

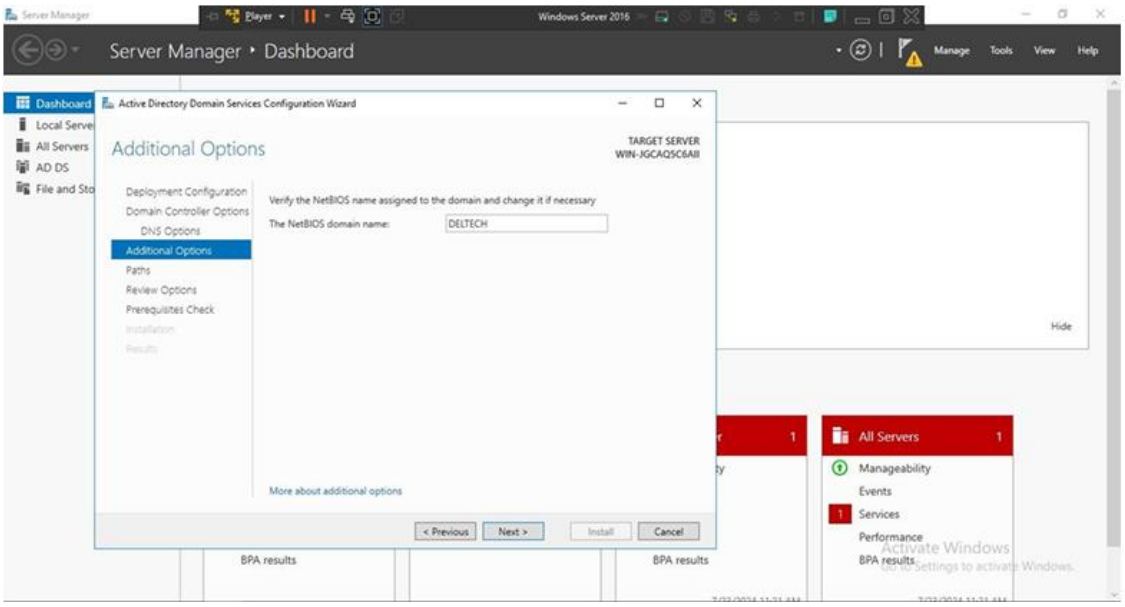

Server Manager
Player
Windows Server 2016
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
File and Storag
File Explorer
Change an Account
Control Panel › User Accounts › User Accounts › Manage Accounts › Change an Account
Search Control Panel
Make changes to Administrator's account
Change the password
Manage another account
Administrator
Local Account
Administrator
Password protected
Hide
Activate Windows
Go to Settings to activate Windows.

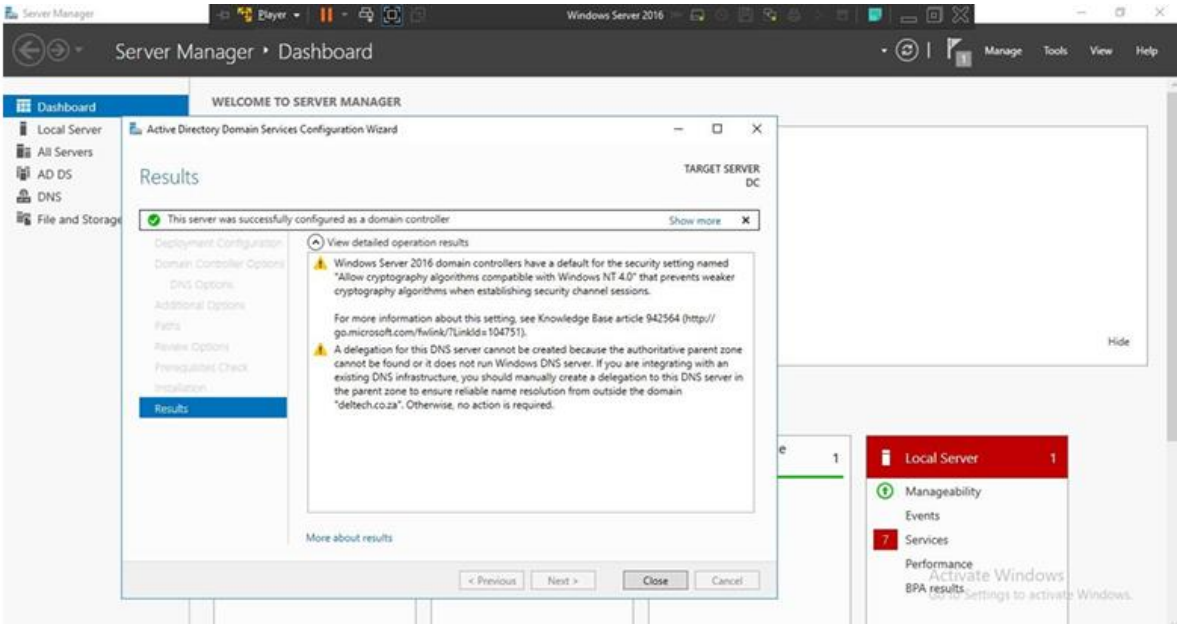

Server Manager
Player
Windows Server 2016
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
File and Storag
File Explorer
Network Connections
Control Panel › Network and Internet › Network Connections
Search Network Connections
Rename this connection
View status of this connection
Change settings of this connection
Ethernet0 Properties
Networking
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 192 . 168 . 10 . 1
Subnet mask: 255 . 255 . 255 . 0
Default gateway: 192 . 168 . 10 . 200
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...
OK
Cancel
1 item
Hide
Activate Windows
Go to Settings to activate Windows.

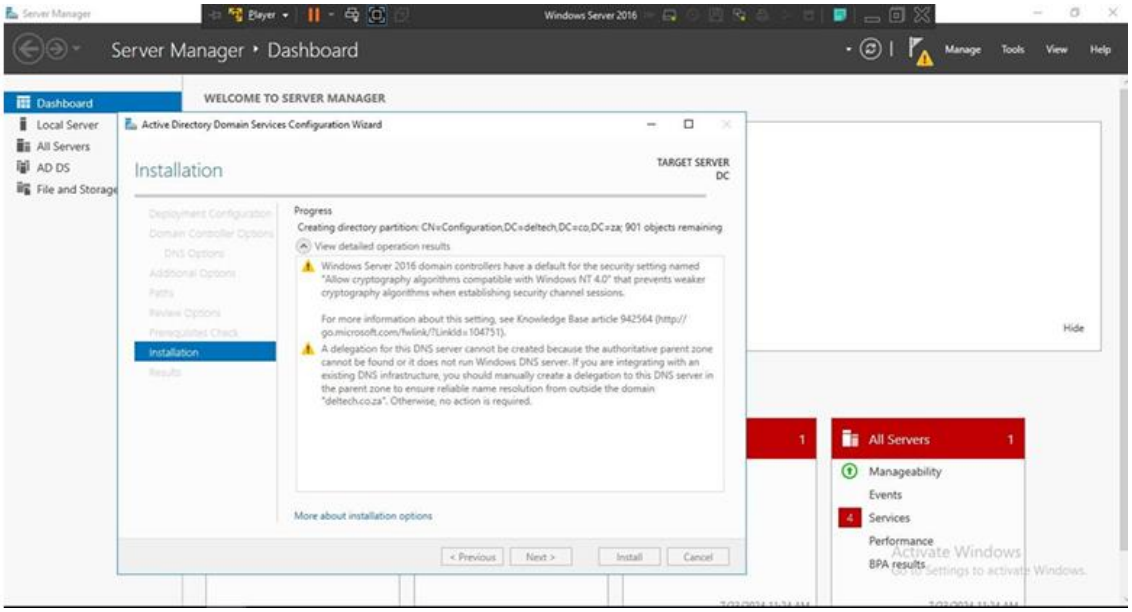

Server Manager
Windows Server 2016
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
File and Storage Services
Add Roles and Features Wizard
Select server roles
DESTINATION SERVER
WIN-JGCAQSC6AII
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Add Roles and Features Wizard
Add features that are required for Active Directory Domain Services?
You cannot install Active Directory Domain Services unless the following role services or features are also installed.
[Tools] Group Policy Management
Remote Server Administration Tools
Role Administration Tools
AD DS and AD LDS Tools
Active Directory module for Windows PowerShell
AD DS Tools
[Tools] Active Directory Administrative Center
[Tools] AD DS Snap-Ins and Command-Line Tools
Include management tools (if applicable)
Add Features
Cancel
< Previous
Next >
Install
Cancel
Hide
Performance
BPA results
Services
Performance
BPA results
Services
Performance
BPA results
Activate Windows
Go to Settings to activate Windows.

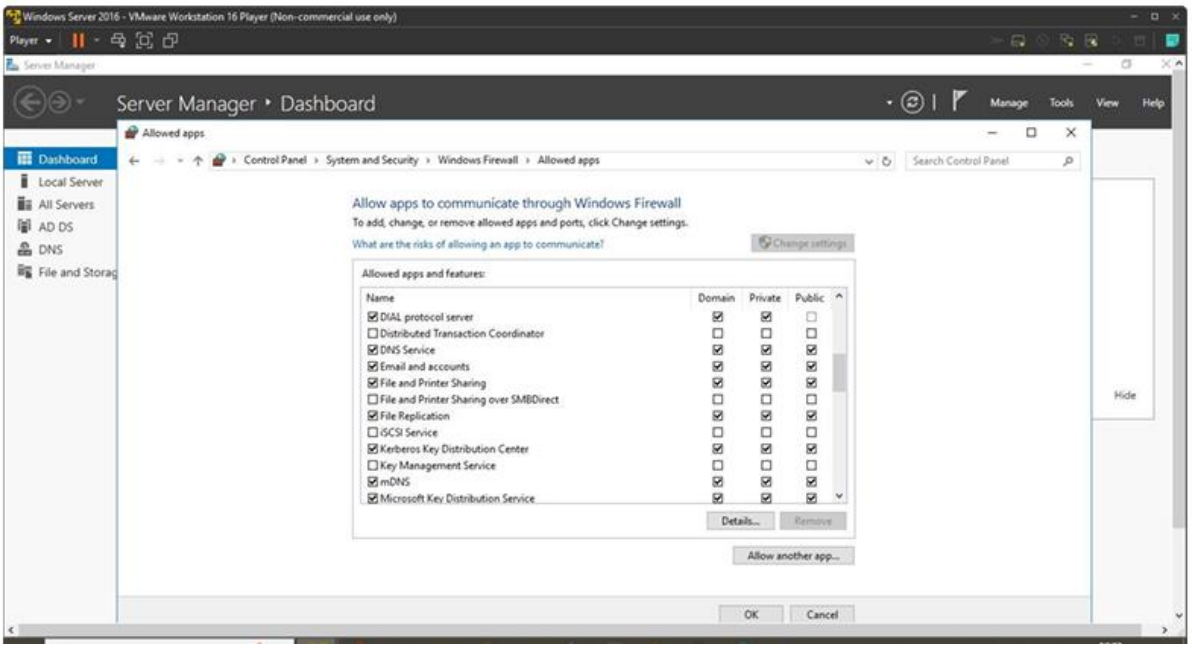

Server Manager
Windows Server 2016
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
File and Storage Services
Add Roles and Features Wizard
Before you begin
DESTINATION SERVER
WIN-JGCAQSC6AII
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
This wizard helps you install roles, role services, or features. You determine which roles, role services, or features to install based on the computing needs of your organization, such as sharing documents, or hosting a website.
To remove roles, role services, or features:
Start the Remove Roles and Features Wizard
Before you continue, verify that the following tasks have been completed:
The Administrator account has a strong password
Network settings, such as static IP addresses, are configured
The most current security updates from Windows Update are installed
If you must verify that any of the preceding prerequisites have been completed, close the wizard, complete the steps, and then run the wizard again.
To continue, click Next.
Skip this page by default
< Previous
Next >
Install
Cancel
Hide
Performance
BPA results
Services
Performance
BPA results
Services
Performance
BPA results
Activate Windows
Go to Settings to activate Windows.


Server Manager
Player
Windows Server 2016
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Dashboard
Local Serve
All Servers
AD DS
File and Sto
Active Directory Domain Services Configuration Wizard
Deployment Configuration
TARGET SERVER
WIN-JGCAQ5C6AII
Deployment Configuration
Domain Controller Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Select the deployment operation
Add a domain controller to an existing domain
Add a new domain to an existing forest
Add a new forest
Specify the domain information for this operation
Root domain name:
deltech.co.za
More about deployment configurations
< Previous
Next >
Install
Cancel
Hide
All Servers
Manageability
Events
Services
Performance
BPA results
Activate Windows
Go to Settings to activate Windows.


Server Manager
Player
Windows Server 2016
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Dashboard
Local Serve
All Servers
AD DS
File and Sto
Active Directory Domain Services Configuration Wizard
Additional Options
TARGET SERVER
WIN-JGCAQ5C6AII
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Verify the NetBIOS name assigned to the domain and change it if necessary
The NetBIOS domain name:
DELTECH
More about additional options
< Previous
Next >
Install
Cancel
Hide
All Servers
Manageability
Events
Services
Performance
BPA results
Activate Windows
Go to Settings to activate Windows.


Server Manager
Player
Windows Server 2016
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD DS
DNS
File and Storage
WELCOME TO SERVER MANAGER
Active Directory Domain Services Configuration Wizard
Results
TARGET SERVER
DC
This server was successfully configured as a domain controller
Show more
View detailed operation results
Windows Server 2016 domain controllers have a default for the security setting named "Allow cryptography algorithms compatible with Windows NT 4.0" that prevents weaker cryptography algorithms when establishing security channel sessions.
For more information about this setting, see Knowledge Base article 942564 (http://go.microsoft.com/fwlink/?LinkId=104751).
A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found or it does not run Windows DNS server. If you are integrating with an existing DNS infrastructure, you should manually create a delegation to this DNS server in the parent zone to ensure reliable name resolution from outside the domain "deltech.co.za". Otherwise, no action is required.
More about results
< Previous
Next >
Close
Cancel
Hide
Local Server
Manageability
Events
Services
Performance
BPA results
Activate Windows

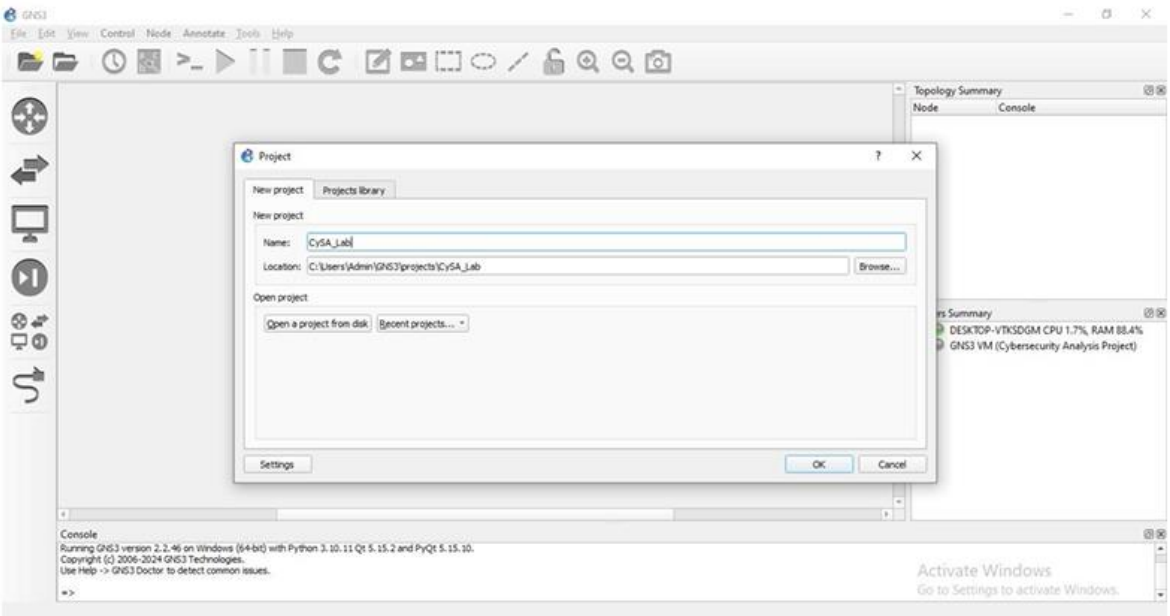

Server Manager
Player
Windows Server 2016
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD DS
File and Storage
WELCOME TO SERVER MANAGER
Active Directory Domain Services Configuration Wizard
Installation
TARGET SERVER
DC
Progress
Creating directory partition: CN=Configuration,DC=deltech,DC=co,DC=za; 901 objects remaining
View detailed operation results
Windows Server 2016 domain controllers have a default for the security setting named "Allow cryptography algorithms compatible with Windows NT 4.0" that prevents weaker cryptography algorithms when establishing security channel sessions.
For more information about this setting, see Knowledge Base article 942564 (http://go.microsoft.com/fwlink/?LinkId=104751).
A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found or it does not run Windows DNS server. If you are integrating with an existing DNS infrastructure, you should manually create a delegation to this DNS server in the parent zone to ensure reliable name resolution from outside the domain "deltech.co.za". Otherwise, no action is required.
More about installation options
< Previous
Next >
Install
Cancel
Hide
All Servers
Manageability
Events
Services
Performance
BPA results
Activate Windows

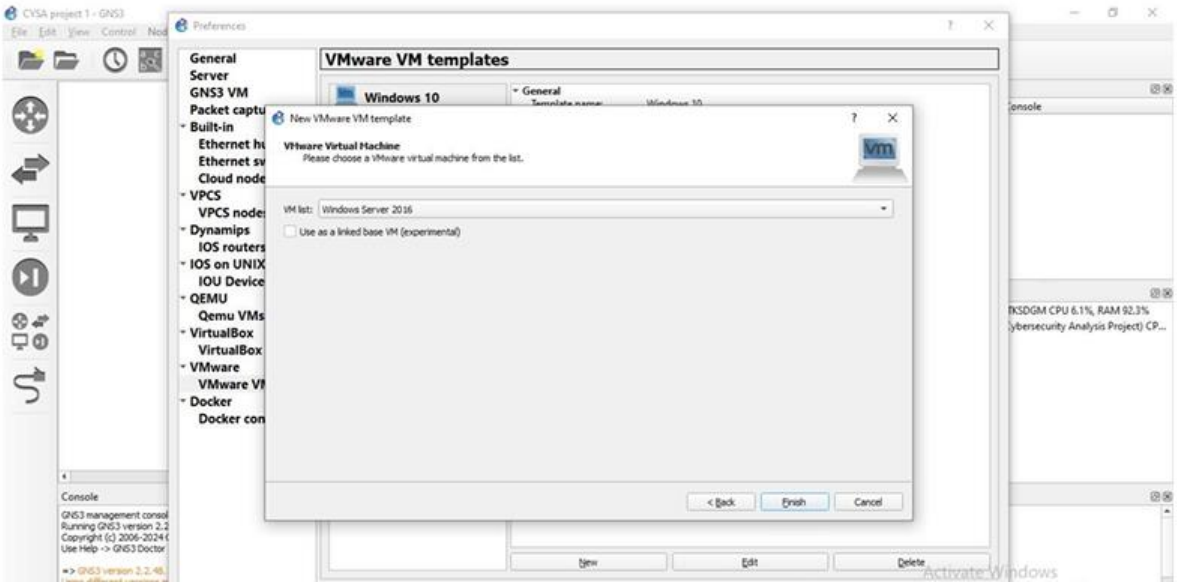


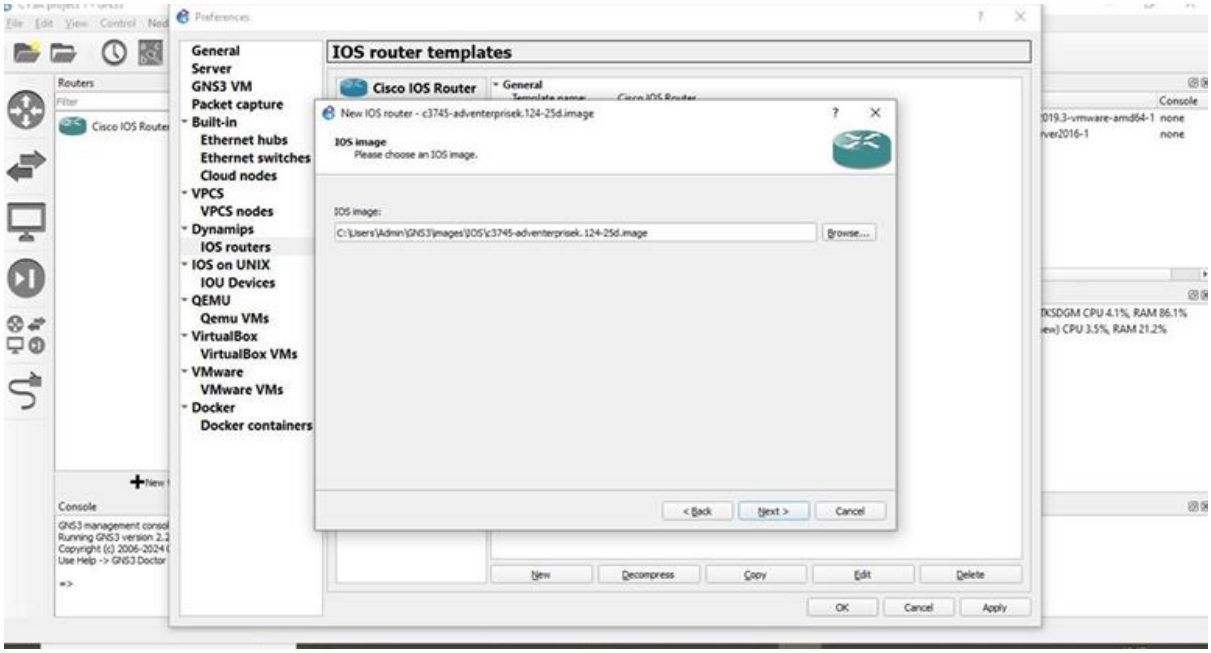
TASK 2:

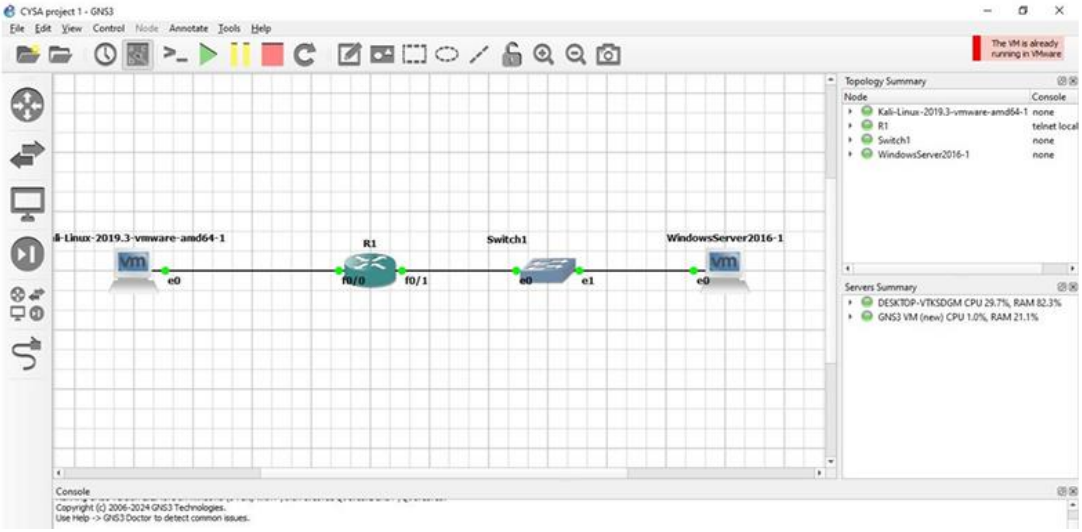
GNS3 LAB SETUP

2.1




Preferences
General
Server
GNS3 VM
Packet captu
Built-in
Ethernet hu
Ethernet sw
Cloud node
VPCS
VPCS node
Dynamips
IOS routers
IOS on UNIX
IOU Device
QEMU
Qemu VMs
VirtualBox
VirtualBox
VMware
VMware VM
Docker
Docker con
VMware VM templates
Windows 10
General
New VMware VM template
VMware Virtual Machine
Please choose a VMware virtual machine from the list.
VM list:
Windows Server 2016
Use as a linked base VM (experimental)
< Back
Finish
Cancel
New
Edit
Delete
Console

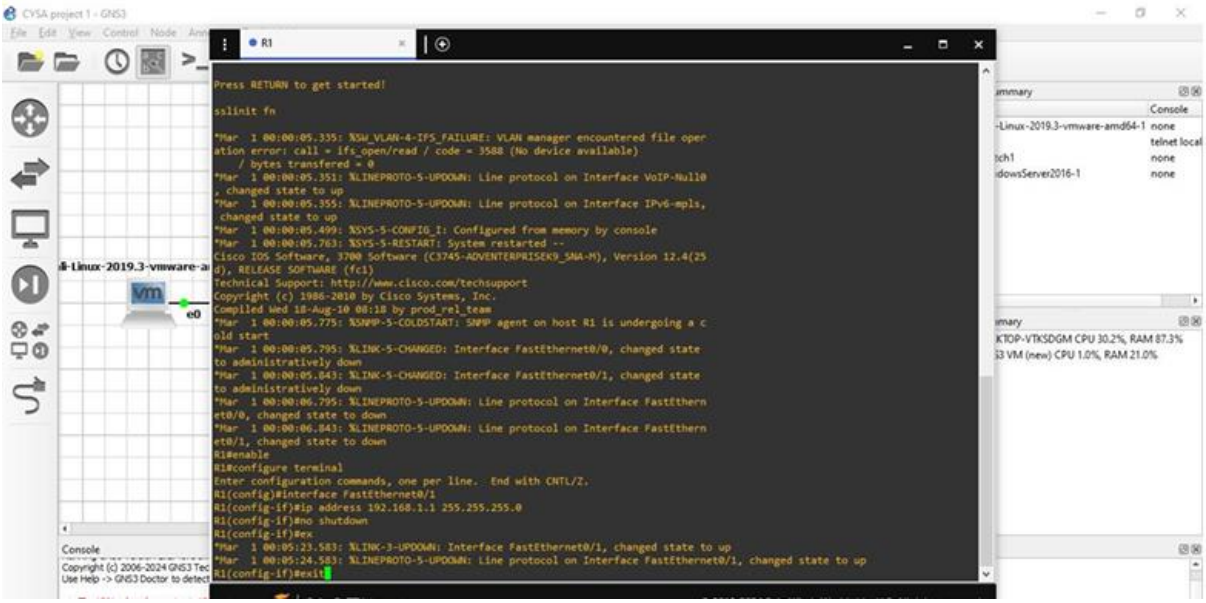

Preferences
General
Server
GNS3 VM
Packet capture
Built-in
Ethernet hubs
Ethernet switches
Cloud nodes
VPCS
VPCS nodes
Dynamips
IOS routers
IOS on UNIX
IOU Devices
QEMU
Qemu VMs
VirtualBox
VirtualBox VMs
VMware
VMware VMs
Docker
Docker containers
IOS router templates
Cisco IOS Router
General
New IOS router - c3745-adventerprisek.124-25d.image
IOS image
Please choose an IOS image.
IOS image:
C:\Users\Admin\GNS3\images\IOS\c3745-adventerprisek.124-25d.image
Browse...
< Back
Next >
Cancel
New
Decompress
Copy
Edit
Delete
OK
Cancel
Apply
Routers
Cisco IOS Router
Console



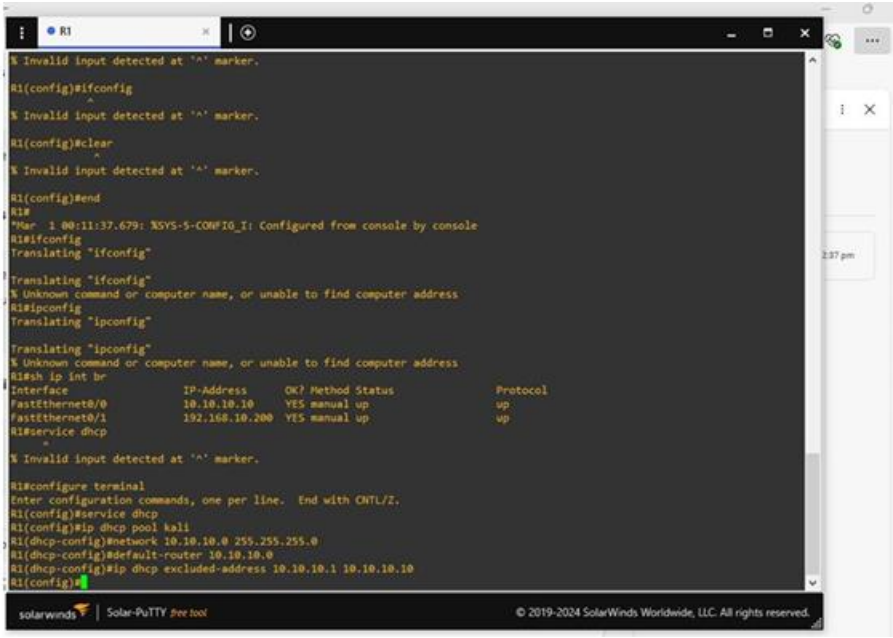
TASK 3:

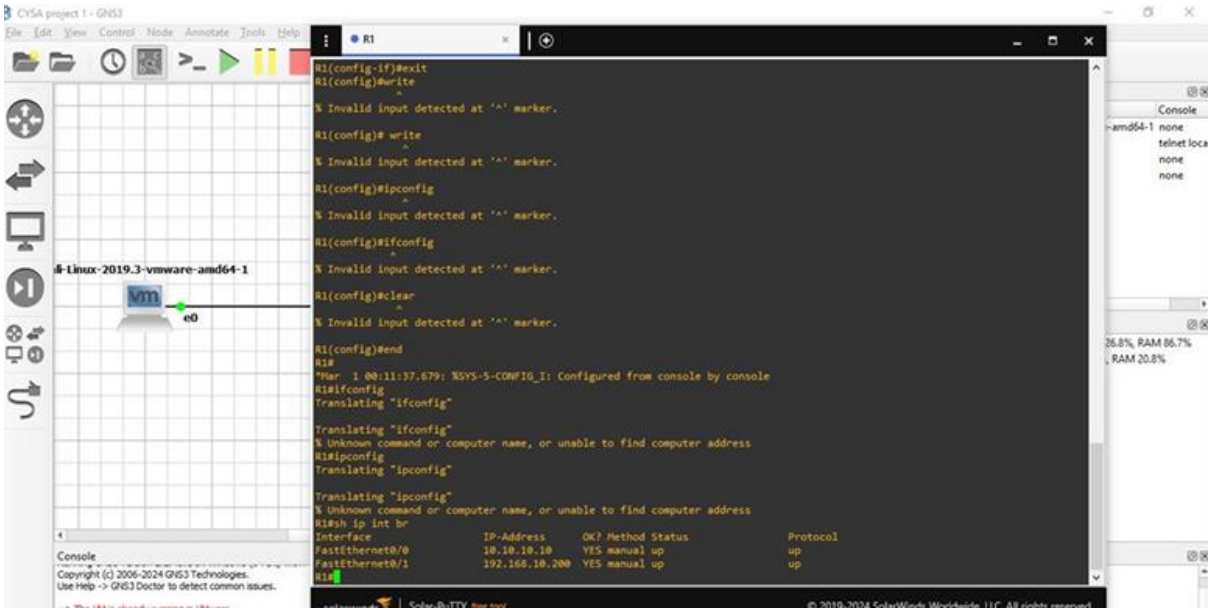
CISCO 3745 ROUTER CONFIGURATION

3.1

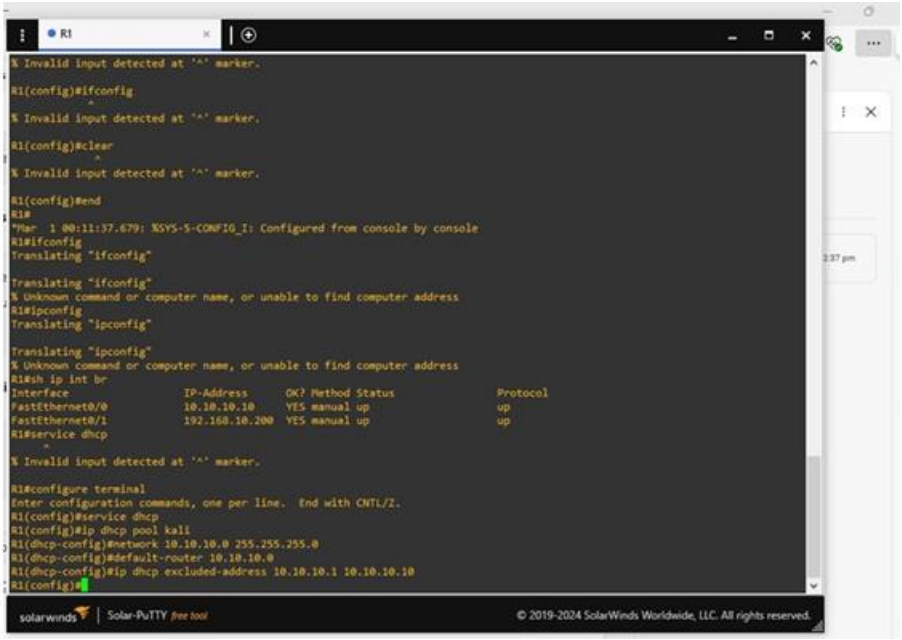


3.2


R1
% Invalid input detected at '^' marker.
R1(config)#ifconfig
% Invalid input detected at '^' marker.
R1(config)#clear
% Invalid input detected at '^' marker.
R1(config)#end
R1#
*Mar 1 00:11:37.679: %SYS-5-CONFIG_I: Configured from console by console
R1#ifconfig
Translating "ifconfig"
Translating "ifconfig"
% Unknown command or computer name, or unable to find computer address
R1#ipconfig
Translating "ipconfig"
Translating "ipconfig"
% Unknown command or computer name, or unable to find computer address
R1#sh ip int br
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.10.10.10 YES manual up up
FastEthernet0/1 192.168.10.200 YES manual up up
R1#service dhcp
% Invalid input detected at '^' marker.
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#service dhcp
R1(config)#ip dhcp pool kali
R1(dhcp-config)#network 10.10.10.0 255.255.255.0
R1(dhcp-config)#default-router 10.10.10.0
R1(dhcp-config)#ip dhcp excluded-address 10.10.10.1 10.10.10.10
R1(config)#
solarwinds
Solar-PuTTY free tool
© 2019-2024 SolarWinds Worldwide, LLC. All rights reserved.


CYSA project 1 - GNS3
File Edit View Control Node Annotate Tools Help
R1
R1(config-if)#exit
R1(config)#write
% Invalid input detected at '^' marker.
R1(config)# write
% Invalid input detected at '^' marker.
R1(config)#ipconfig
% Invalid input detected at '^' marker.
R1(config)#ifconfig
% Invalid input detected at '^' marker.
R1(config)#clear
% Invalid input detected at '^' marker.
R1(config)#end
R1#
*Mar 1 00:11:37.679: %SYS-5-CONFIG_I: Configured from console by console
R1#ifconfig
Translating "ifconfig"
Translating "ifconfig"
% Unknown command or computer name, or unable to find computer address
R1#ipconfig
Translating "ipconfig"
Translating "ipconfig"
% Unknown command or computer name, or unable to find computer address
R1#sh ip int br
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.10.10.10 YES manual up up
FastEthernet0/1 192.168.10.200 YES manual up up
R1#
Console
Copyright (c) 2006-2024 GNS3 Technologies.
Use Help -> GNS3 Doctor to detect common issues.
vm
e0

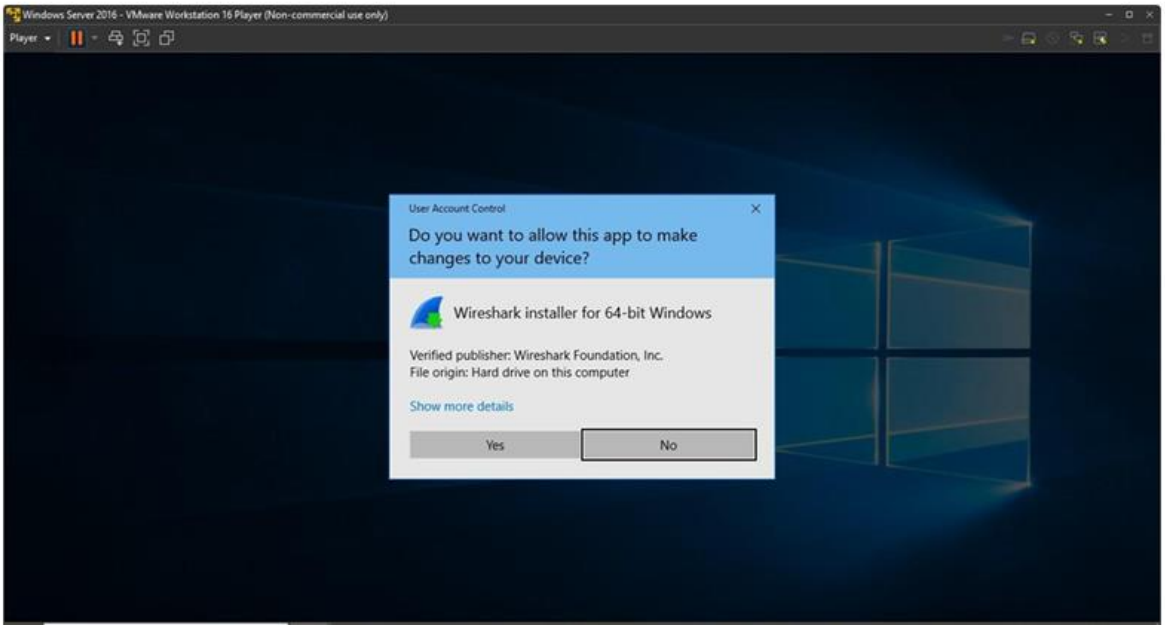
3.3

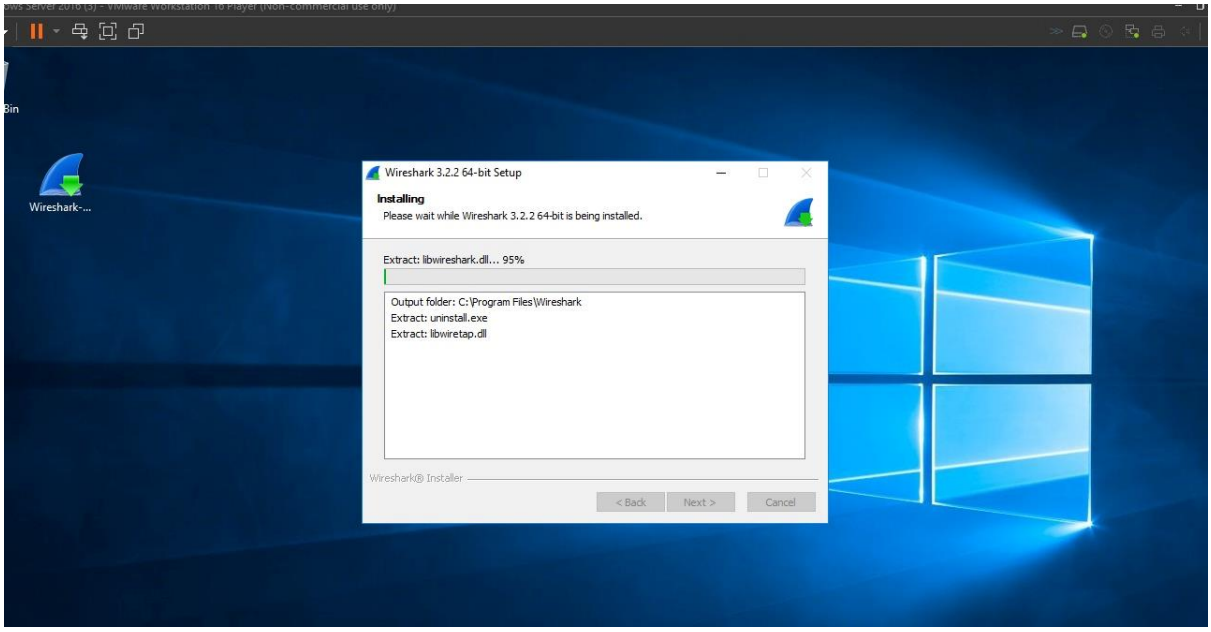


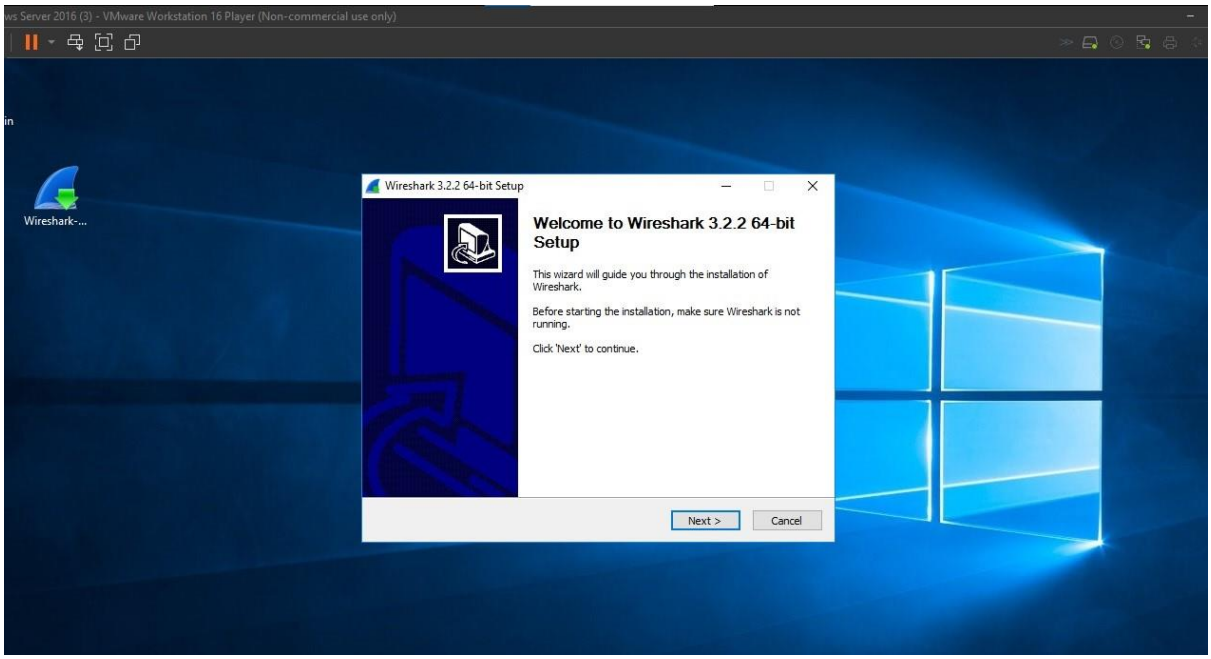
SECTION B TASK 4:

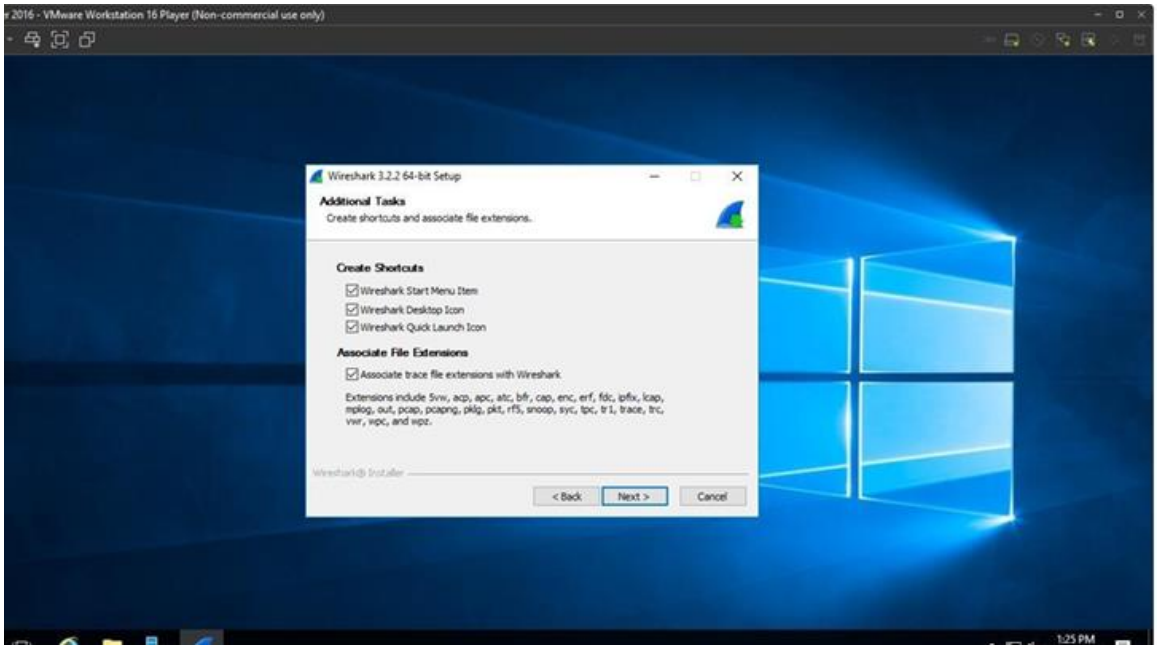
PERFORMING RECONNAISSANCE

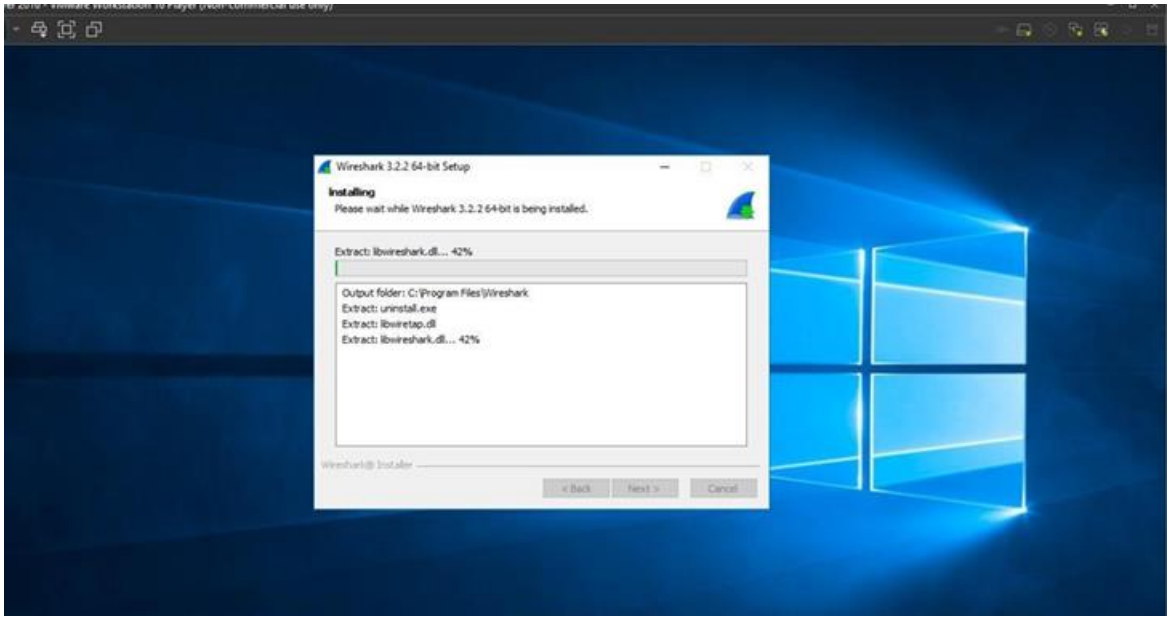
4.1

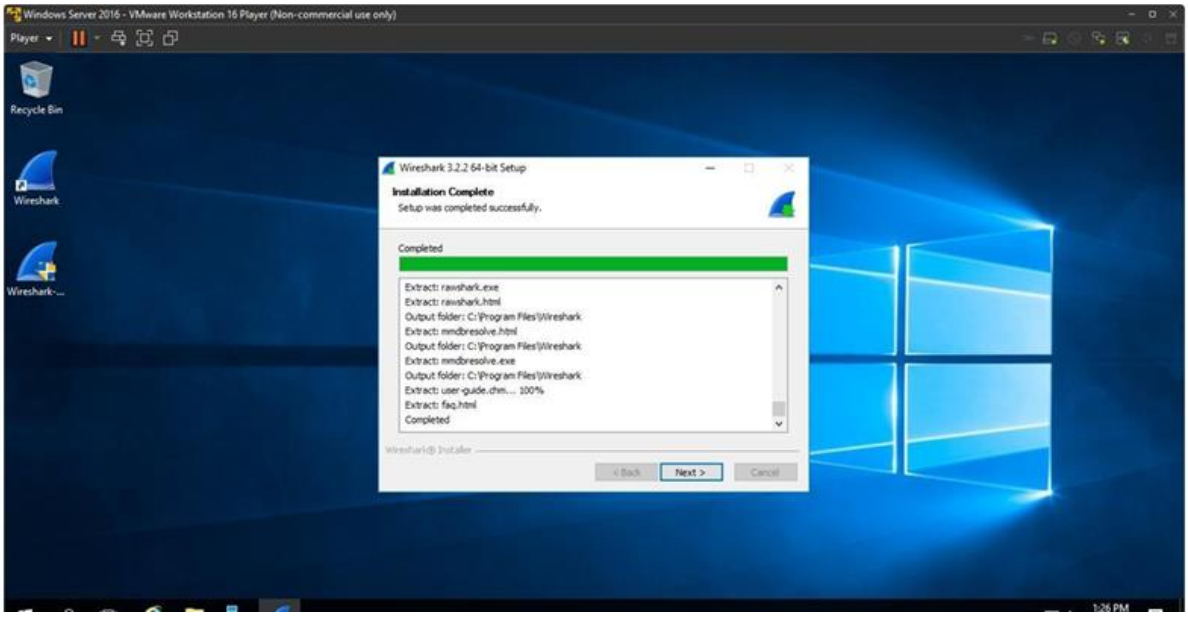



Wireshark-...
Wireshark 3.2.2 64-bit Setup
Installing
Please wait while Wireshark 3.2.2 64-bit is being installed.
Extract: libwireshark.dll... 95%
Output folder: C:\Program Files\Wireshark
Extract: uninstall.exe
Extract: libwiretap.dll
Wireshark® Installer
< Back
Next >
Cancel

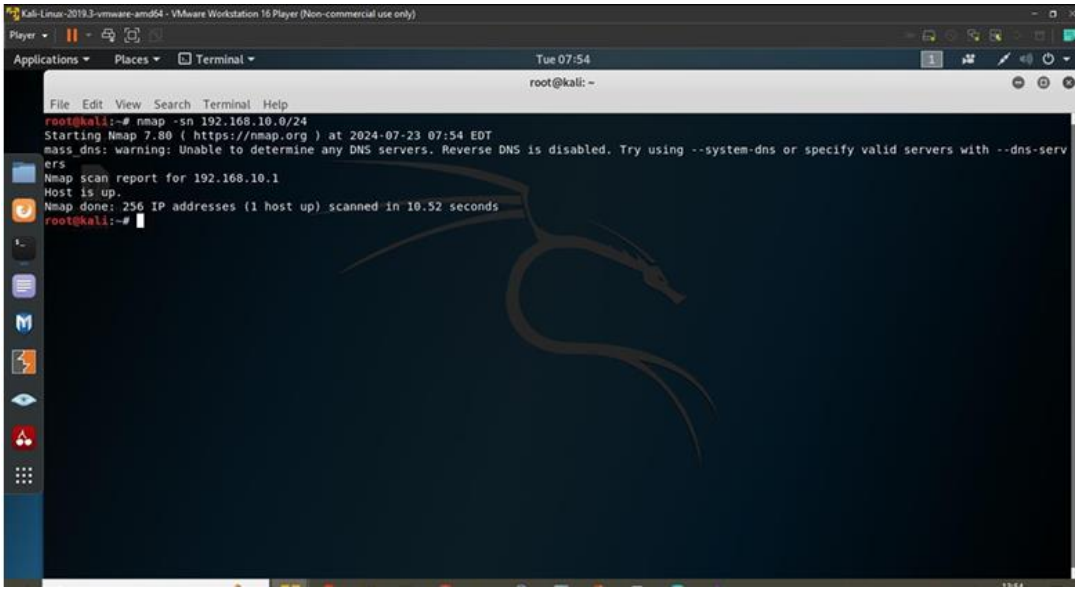

Wireshark-...
Wireshark 3.2.2 64-bit Setup
Welcome to Wireshark 3.2.2 64-bit Setup
This wizard will guide you through the installation of Wireshark.
Before starting the installation, make sure Wireshark is not running.
Click 'Next' to continue.
Next >
Cancel

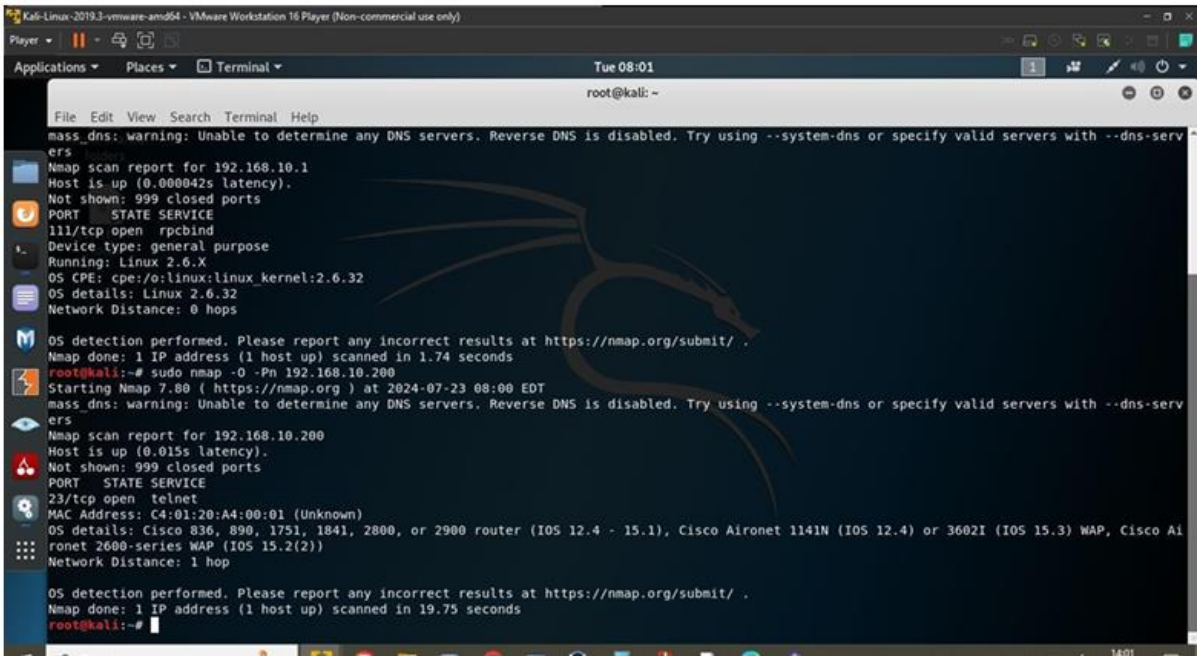

2016 - VMware Workstation 16 Player (Non-commercial use only)
Wireshark 3.2.2 64-bit Setup
Additional Tasks
Create shortcuts and associate file extensions.
Create Shortcuts
Wireshark Start Menu Item
Wireshark Desktop Icon
Wireshark Quick Launch Icon
Associate File Extensions
Associate trace file extensions with Wireshark
Extensions include 5vw, acp, apc, atc, bfr, cap, enc, erf, fdc, ipfix, lcap, mplog, out, pcap, pcapng, pklg, pkt, rf5, snoop, syc, tpc, tr1, trace, trc, vwr, wpc, and wpz.
Wireshark® Installer
< Back
Next >
Cancel
1:25 PM


Wireshark 3.2.2 64-bit Setup
Installing
Please wait while Wireshark 3.2.2 64-bit is being installed.
Extract: libwireshark.dll... 42%
Output folder: C:\Program Files\Wireshark
Extract: uninstall.exe
Extract: libwiretap.dll
Extract: libwireshark.dll... 42%
Wireshark® Installer
< Back
Next >
Cancel


Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Player
Recycle Bin
Wireshark
Wireshark-...
Wireshark 3.2.2 64-bit Setup
Installation Complete
Setup was completed successfully.
Completed
Extract: rawshark.exe
Extract: rawshark.html
Output folder: C:\Program Files\Wireshark
Extract: mmdbresolve.html
Output folder: C:\Program Files\Wireshark
Extract: mmdbresolve.exe
Output folder: C:\Program Files\Wireshark
Extract: user-guide.chm... 100%
Extract: faq.html
Completed
Wireshark® Installer
< Back
Next >
Cancel

4.2

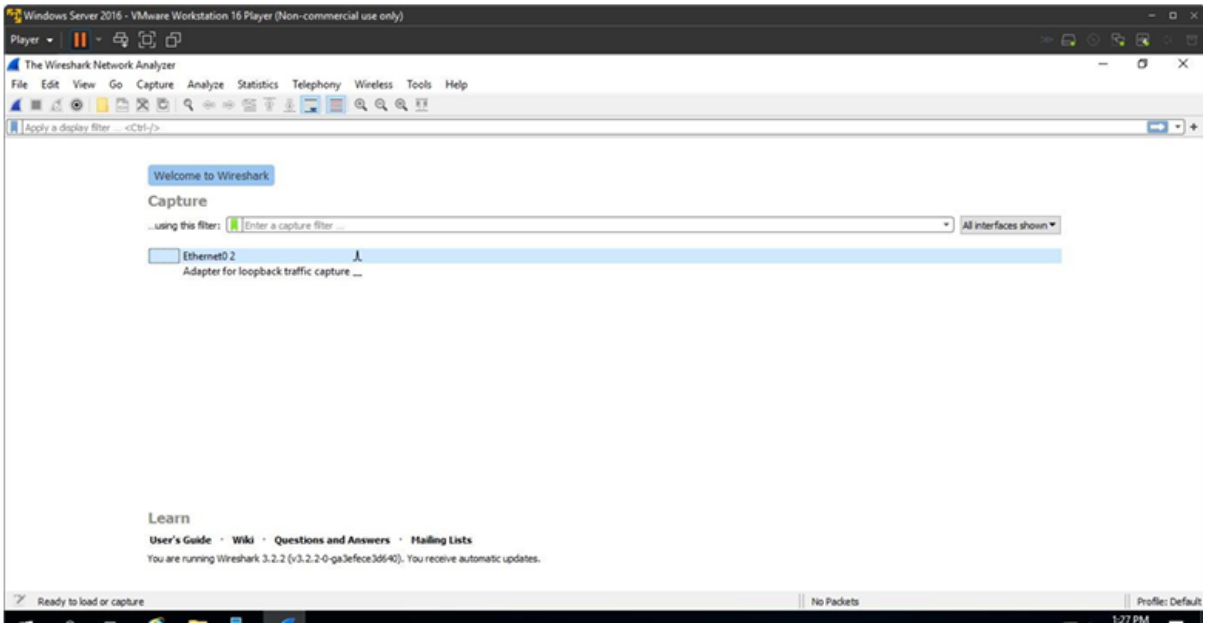

Kali-Linux-2019.3-vmware-amd64 - VMware Workstation 16 Player (Non-commercial use only)
Player
Applications
Places
Terminal
Tue 07:54
root@kali: ~
File Edit View Search Terminal Help
root@kali:~# nmap -sn 192.168.10.0/24
Starting Nmap 7.80 (https://nmap.org) at 2024-07-23 07:54 EDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.10.1
Host is up.
Nmap done: 256 IP addresses (1 host up) scanned in 10.52 seconds
root@kali:~#

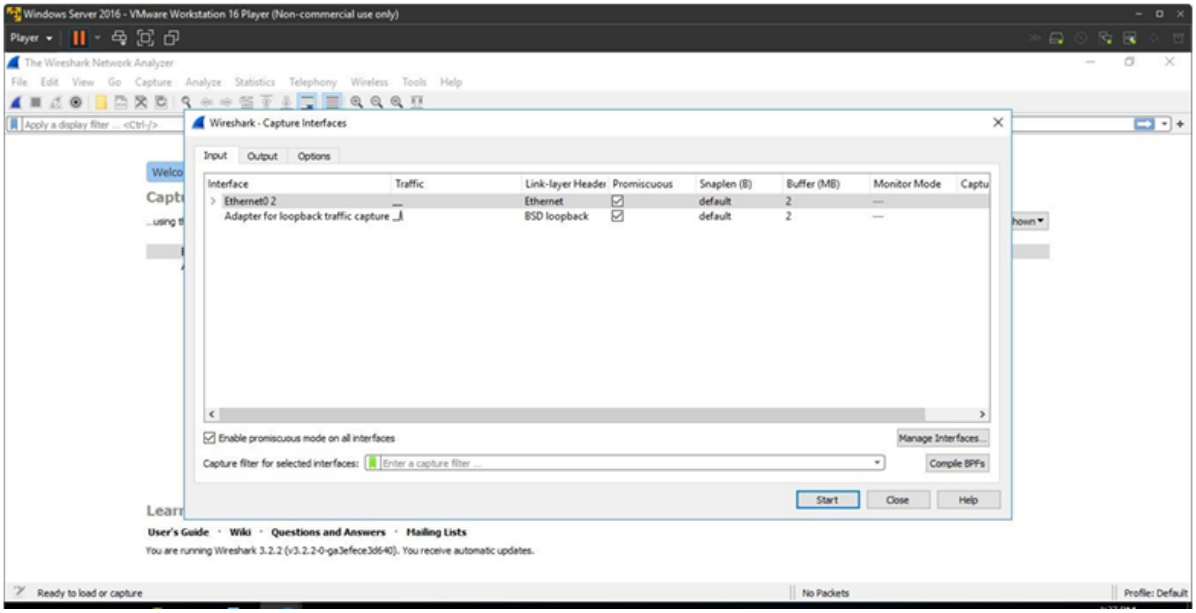


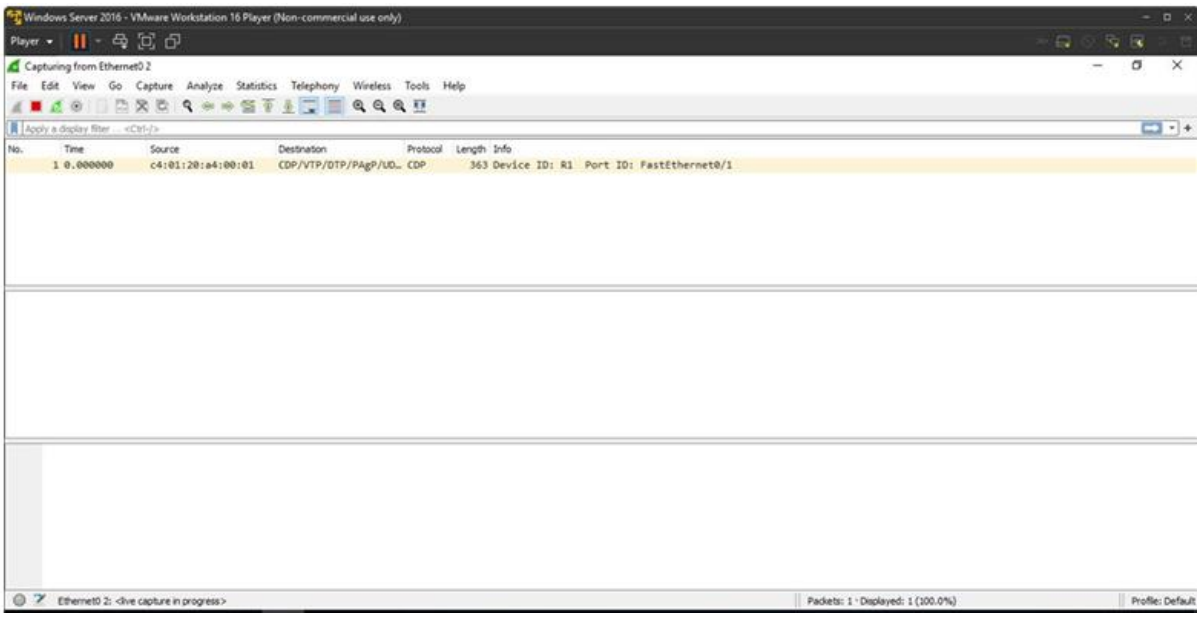
SECTION C TASK 5:

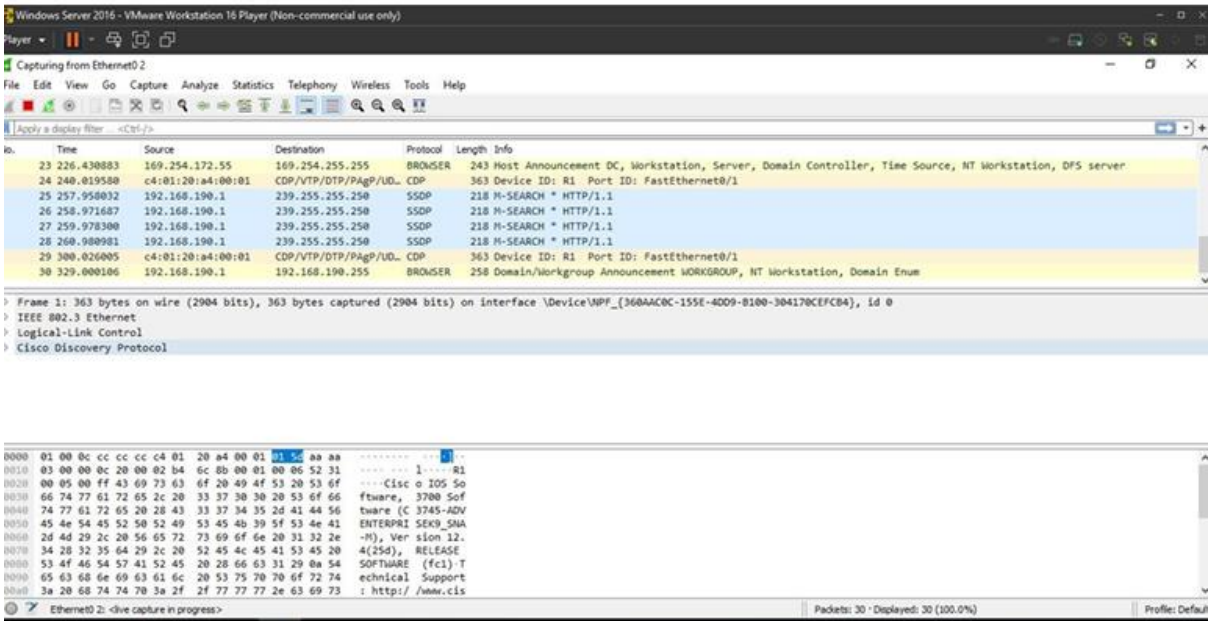
ANALYSING DATA WITH WIRESHARK

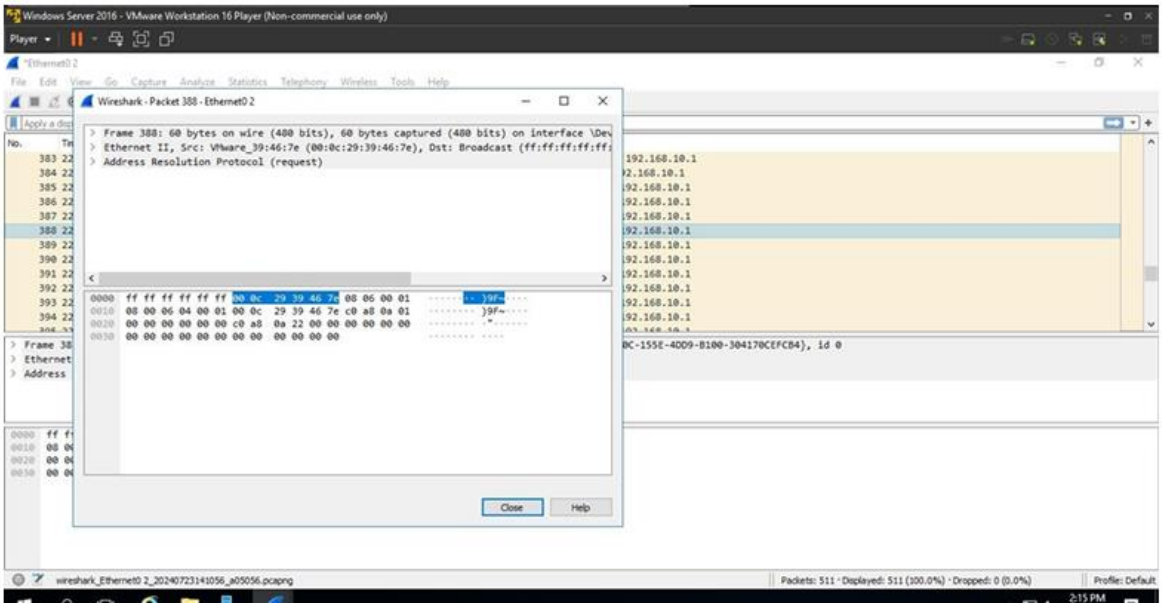
5.1

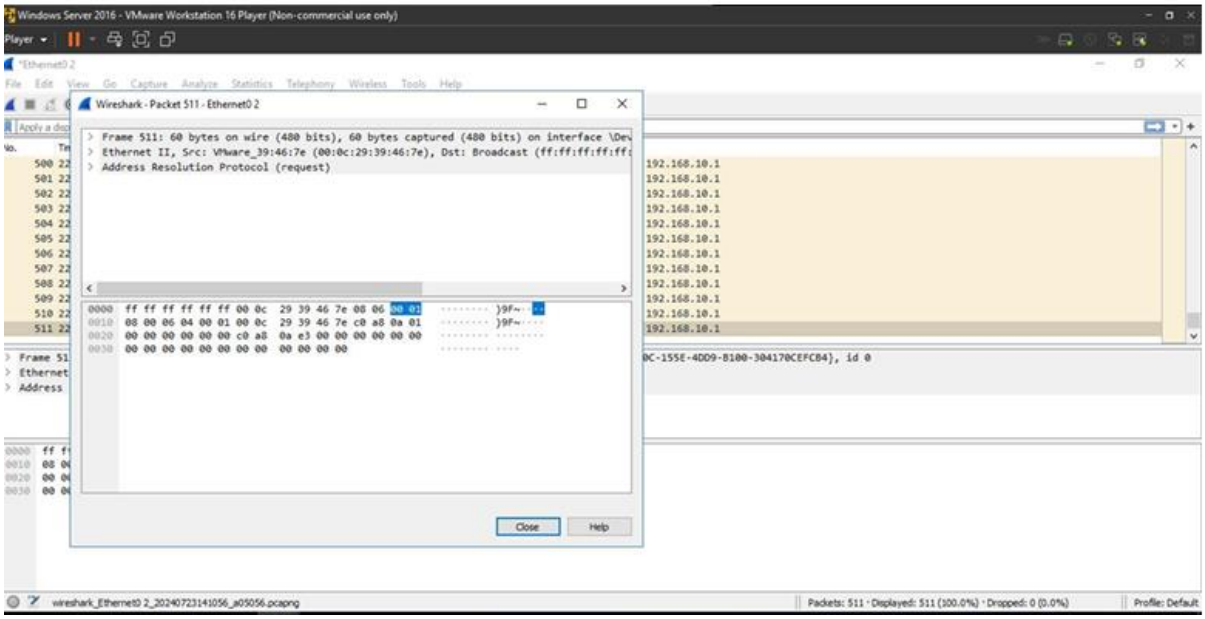



Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Player
The Wireshark Network Analyzer
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
Wireshark · Capture Interfaces
Input Output Options
Interface Traffic Link-layer Header Promiscuous Snaplen (B) Buffer (MB) Monitor Mode
Ethernet0 2 Ethernet default 2
Adapter for loopback traffic capture BSD loopback default 2
Enable promiscuous mode on all interfaces
Manage Interfaces...
Capture filter for selected interfaces:
Enter a capture filter ...
Compile BPFs
Start Close Help
User's Guide · Wiki · Questions and Answers · Mailing Lists
You are running Wireshark 3.2.2 (v3.2.2-0-ga3efece3d640). You receive automatic updates.
Ready to load or capture
No Packets
Profile: Default


Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Player
Capturing from Ethernet0 2
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000 c4:01:20:a4:00:01 CDP/VTP/DTP/PAgP/UD... CDP 363 Device ID: R1 Port ID: FastEthernet0/1
Ethernet0 2: <live capture in progress>
Packets: 1 · Displayed: 1 (100.0%)
Profile: Default


Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Capturing from Ethernet0 2
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
23 226.430883 169.254.172.55 169.254.255.255 BROWSER 243 Host Announcement DC, Workstation, Server, Domain Controller, Time Source, NT Workstation, DFS server
24 240.019580 c4:01:20:a4:00:01 CDP/VTP/DTP/PAgP/UD… CDP 363 Device ID: R1 Port ID: FastEthernet0/1
25 257.950032 192.168.190.1 239.255.255.250 SSDP 218 M-SEARCH * HTTP/1.1
26 258.971687 192.168.190.1 239.255.255.250 SSDP 218 M-SEARCH * HTTP/1.1
27 259.978300 192.168.190.1 239.255.255.250 SSDP 218 M-SEARCH * HTTP/1.1
28 260.980981 192.168.190.1 239.255.255.250 SSDP 218 M-SEARCH * HTTP/1.1
29 300.026005 c4:01:20:a4:00:01 CDP/VTP/DTP/PAgP/UD… CDP 363 Device ID: R1 Port ID: FastEthernet0/1
30 329.000106 192.168.190.1 192.168.190.255 BROWSER 258 Domain/Workgroup Announcement WORKGROUP, NT Workstation, Domain Enum
Frame 1: 363 bytes on wire (2904 bits), 363 bytes captured (2904 bits) on interface \Device\NPF_{360AAC0C-155E-4DD9-B100-304170CEFCB4}, id 0
IEEE 802.3 Ethernet
Logical-Link Control
Cisco Discovery Protocol
Ethernet0 2: <live capture in progress>
Packets: 30 · Displayed: 30 (100.0%)
Profile: Default


Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Wireshark - Packet 388 - Ethernet0 2
Frame 388: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Dev
Ethernet II, Src: VMware_39:46:7e (00:0c:29:39:46:7e), Dst: Broadcast (ff:ff:ff:ff:ff:
Address Resolution Protocol (request)
Close
Help
wireshark_Ethernet0 2_20240723141056_a05056.pcapng
Packets: 511 · Displayed: 511 (100.0%) · Dropped: 0 (0.0%)
Profile: Default



5.4. SYN

This flag is the one that controls flags in the TCP header, and it's used to initiate TCP connections between two systems in a process called the TCP three-way handshake.

b. ACK

This flag is also a control flag in the TCP header, and it's used to acknowledge the receipt of a packet or segment and its basically and essential part in maintaining reliability in TCP communications.

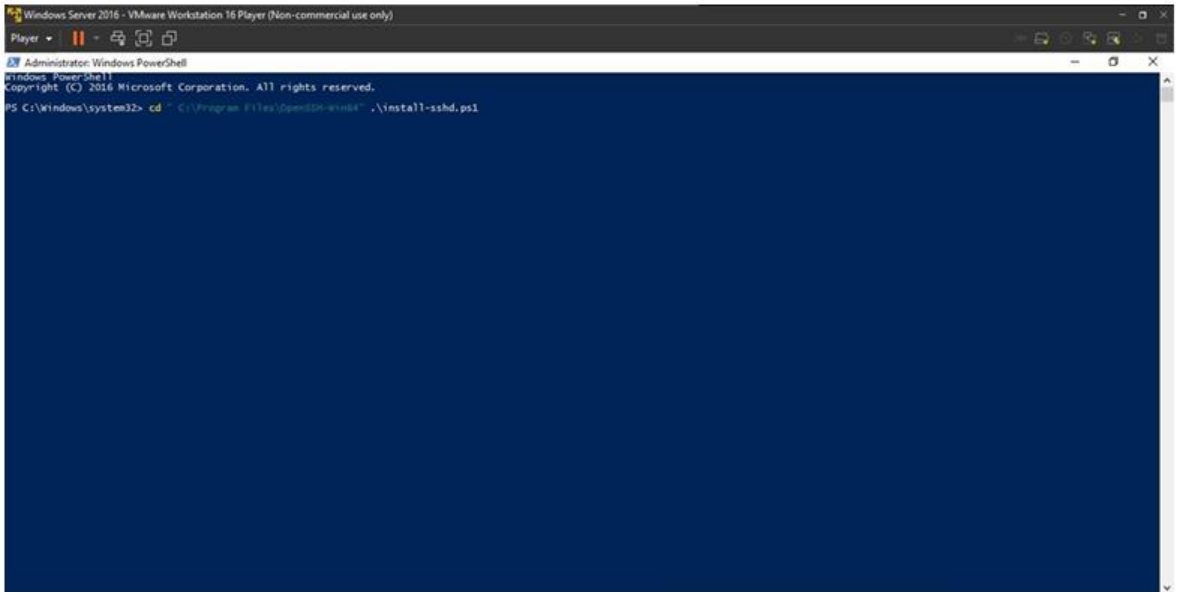
c. RST

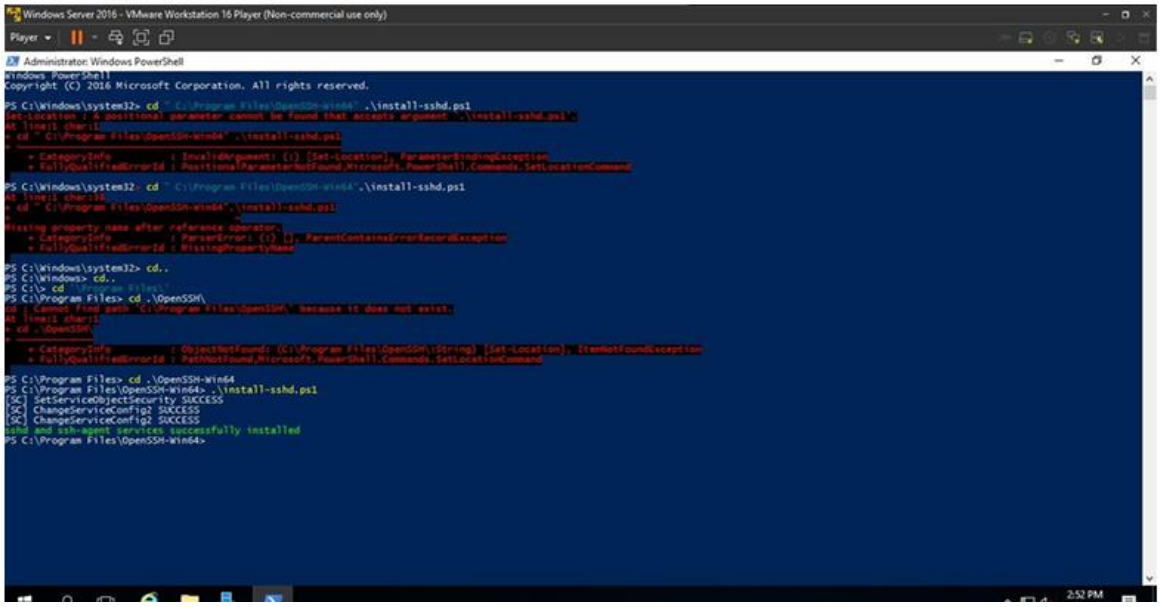
This flag is used in the TCP to abruptly terminate an active connection it signals that something has gone wrong or that the server wants to reset the connection immediately, a TCP connection can be reset by sending RST packet in response to an unexpected packet.

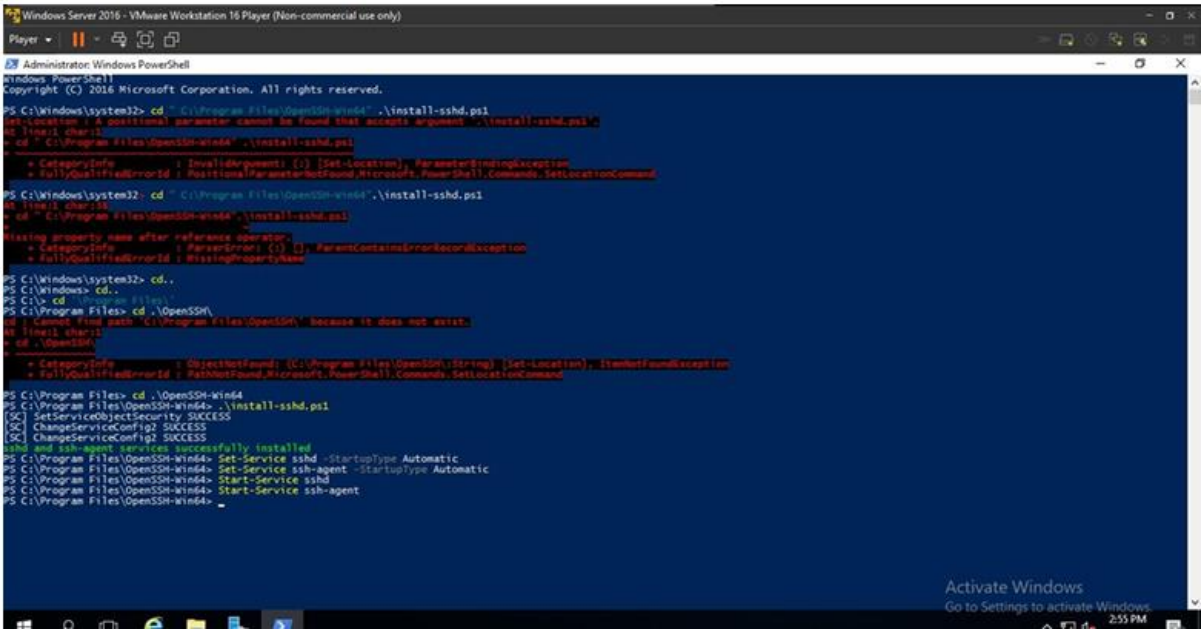
SECTION D TASK 6:

SSH HACKING

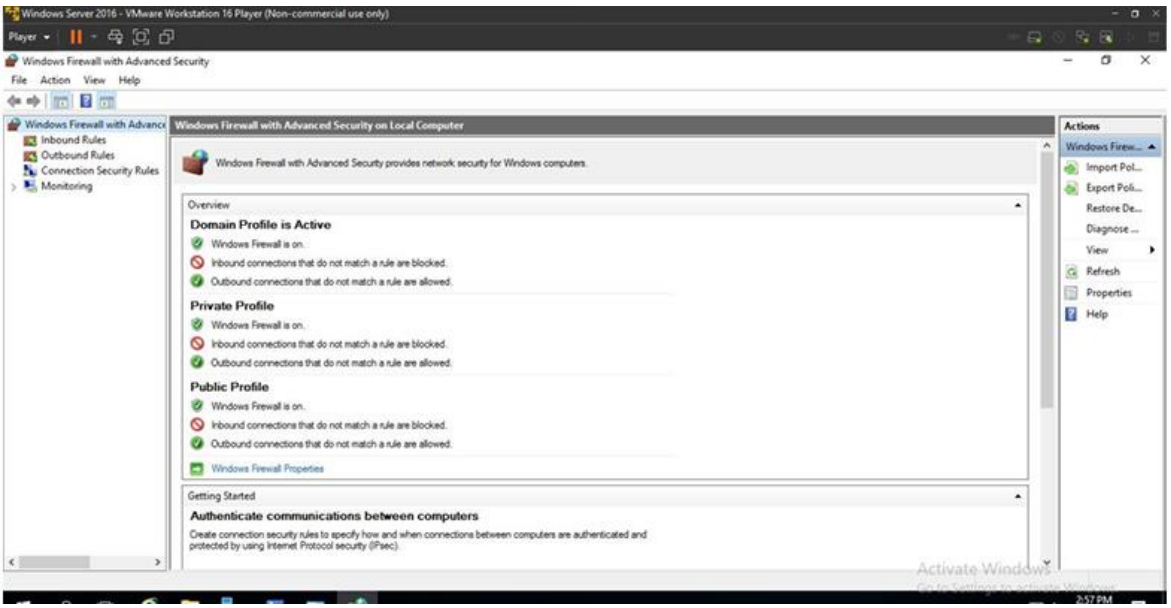
6.1

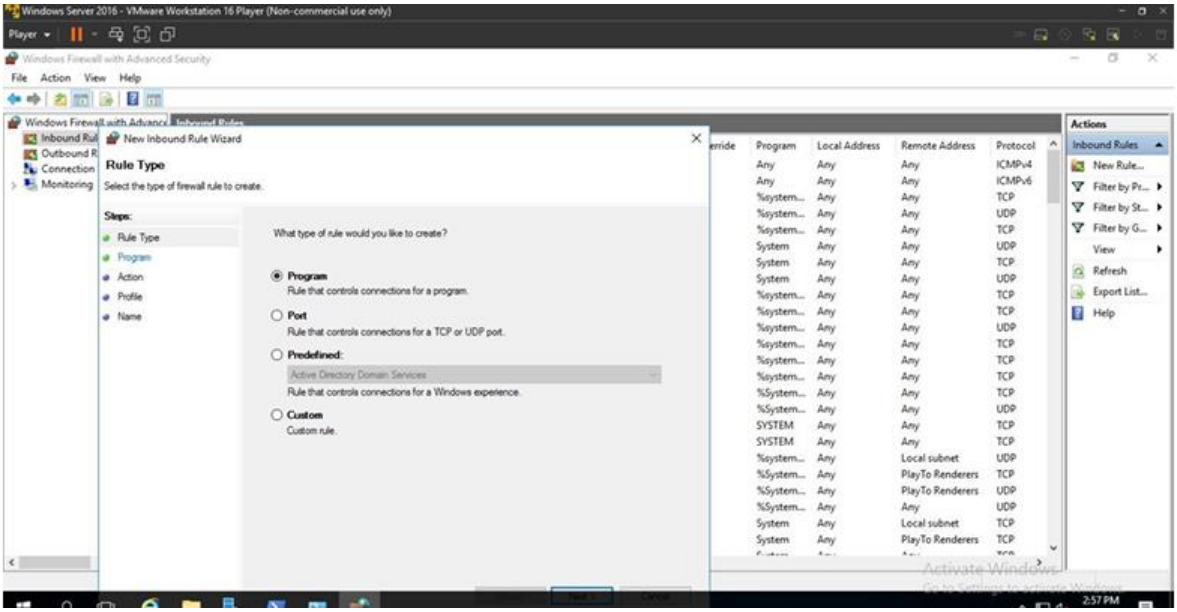


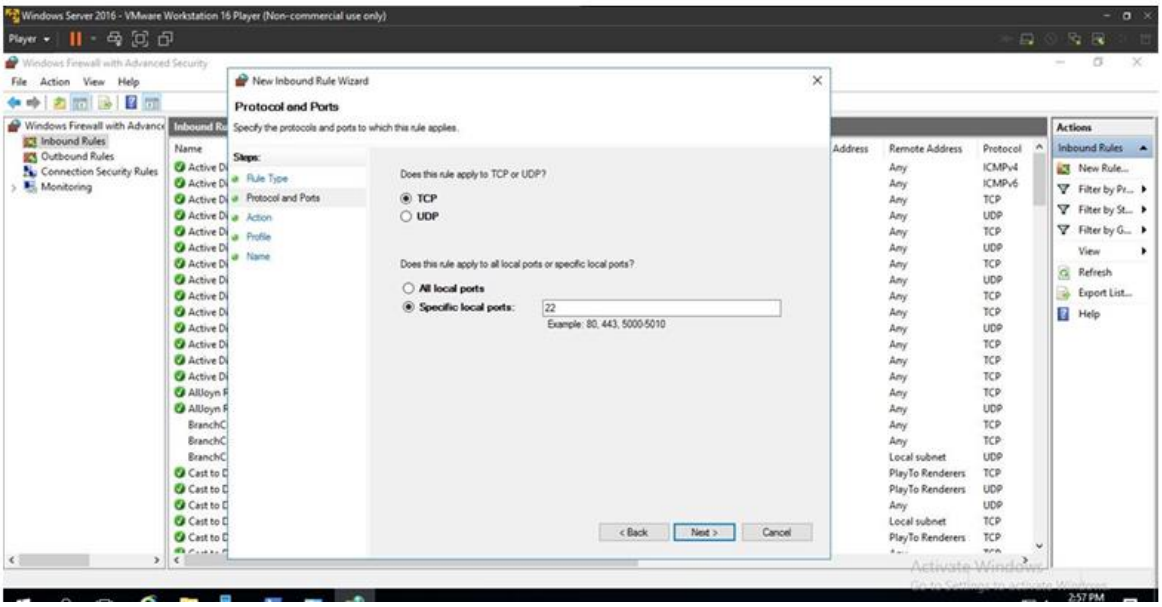

Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Player
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) 2016 Microsoft Corporation. All rights reserved.
PS C:\Windows\system32> cd " C:\Program Files\OpenSSH-Win64" .\install-sshd.ps1
Set-Location : A positional parameter cannot be found that accepts argument ".\install-sshd.ps1".
At line:1 char:1
+ cd " C:\Program Files\OpenSSH-Win64" .\install-sshd.ps1
+ CategoryInfo : InvalidArgument: (:) [Set-Location], ParameterBindingException
+ FullyQualifiedErrorId : PositionalParameterNotFound,Microsoft.PowerShell.Commands.SetLocationCommand
PS C:\Windows\system32> cd " C:\Program Files\OpenSSH-Win64".\install-sshd.ps1
At line:1 char:38
+ cd " C:\Program Files\OpenSSH-Win64".\install-sshd.ps1
+ ~~~

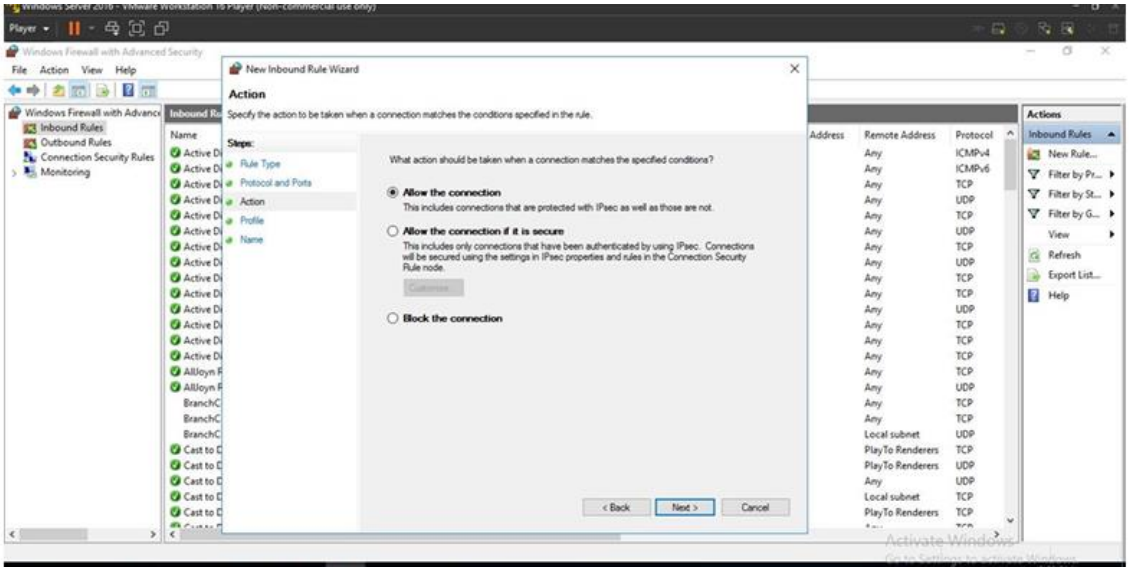

Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) 2016 Microsoft Corporation. All rights reserved.
PS C:\Windows\system32> cd " C:\Program Files\OpenSSH-Win64",.\install-sshd.ps1
PS C:\Windows\system32> cd " C:\Program Files\OpenSSH-Win64",.\install-sshd.ps1
PS C:\Windows\system32> cd..
PS C:\Windows> cd..
PS C:\> cd "\Program Files\"
PS C:\Program Files> cd .\OpenSSH\
PS C:\Program Files> cd .\OpenSSH-Win64
PS C:\Program Files\OpenSSH-Win64> .\install-sshd.ps1
[SC] SetServiceObjectSecurity SUCCESS
[SC] ChangeServiceConfig2 SUCCESS
[SC] ChangeServiceConfig2 SUCCESS
sshd and ssh-agent services successfully installed
PS C:\Program Files\OpenSSH-Win64> Set-Service sshd -StartupType Automatic
PS C:\Program Files\OpenSSH-Win64> Set-Service ssh-agent -StartupType Automatic
PS C:\Program Files\OpenSSH-Win64> Start-Service sshd
PS C:\Program Files\OpenSSH-Win64> Start-Service ssh-agent
PS C:\Program Files\OpenSSH-Win64>
Activate Windows
Go to Settings to activate Windows.
2:55 PM

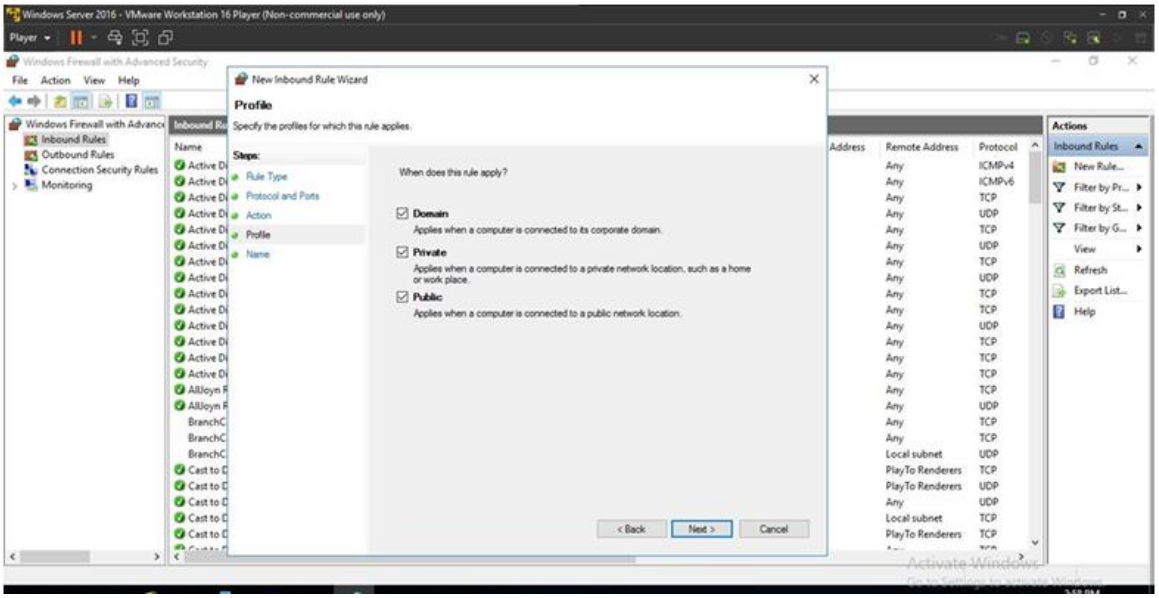
6.2


Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Windows Firewall with Advanced Security
File Action View Help
Windows Firewall with Advanc
Inbound Rules
Outbound Rules
Connection Security Rules
Monitoring
Windows Firewall with Advanced Security on Local Computer
Windows Firewall with Advanced Security provides network security for Windows computers.
Overview
Domain Profile is Active
Windows Firewall is on.
Inbound connections that do not match a rule are blocked.
Outbound connections that do not match a rule are allowed.
Private Profile
Windows Firewall is on.
Inbound connections that do not match a rule are blocked.
Outbound connections that do not match a rule are allowed.
Public Profile
Windows Firewall is on.
Inbound connections that do not match a rule are blocked.
Outbound connections that do not match a rule are allowed.
Windows Firewall Properties
Getting Started
Authenticate communications between computers
Create connection security rules to specify how and when connections between computers are authenticated and protected by using Internet Protocol security (IPsec).
Actions
Windows Firew...
Import Pol...
Export Poli...
Restore De...
Diagnose ...
View
Refresh
Properties
Help
Activate Windows
2:57 PM

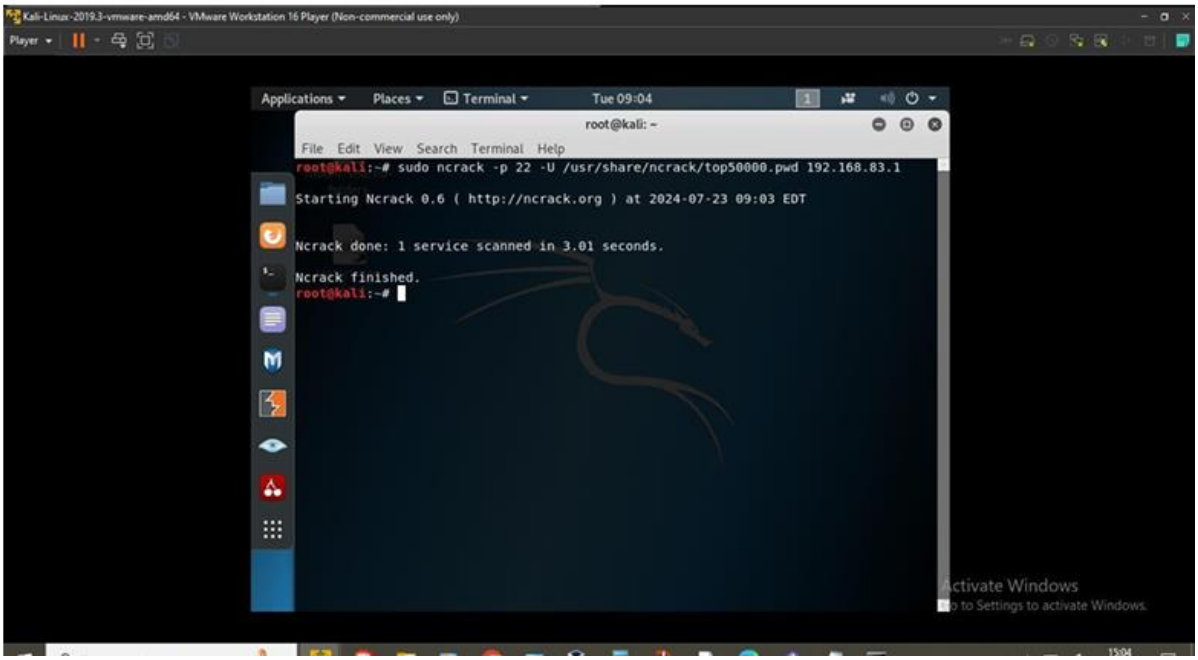

Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Windows Firewall with Advanced Security
File Action View Help
New Inbound Rule Wizard
Rule Type
Select the type of firewall rule to create.
Steps:
Rule Type
Program
Action
Profile
Name
What type of rule would you like to create?
Program
Rule that controls connections for a program.
Port
Rule that controls connections for a TCP or UDP port.
Predefined:
Active Directory Domain Services
Rule that controls connections for a Windows experience.
Custom
Custom rule.
Actions
Inbound Rules
New Rule...
Refresh
Export List...
Help


Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Windows Firewall with Advanced Security
File Action View Help
Inbound Rules
Outbound Rules
Connection Security Rules
Monitoring
New Inbound Rule Wizard
Protocol and Ports
Specify the protocols and ports to which this rule applies.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
Does this rule apply to TCP or UDP?
TCP
UDP
Does this rule apply to all local ports or specific local ports?
All local ports
Specific local ports:
22
Example: 80, 443, 5000-5010
< Back
Next >
Cancel
Actions
Inbound Rules
New Rule...
Refresh
Export List...
Help


Windows Firewall with Advanced Security
File Action View Help
Windows Firewall with Advanc
Inbound Rules
Outbound Rules
Connection Security Rules
Monitoring
New Inbound Rule Wizard
Action
Specify the action to be taken when a connection matches the conditions specified in the rule.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
What action should be taken when a connection matches the specified conditions?
Allow the connection
This includes connections that are protected with IPsec as well as those are not.
Allow the connection if it is secure
This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.
Customize...
Block the connection
< Back
Next >
Cancel
Actions
Inbound Rules
New Rule...
Filter by Pr...
Filter by St...
Filter by G...
View
Refresh
Export List...
Help


Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Windows Firewall with Advanced Security
File Action View Help
Windows Firewall with Advanc
Inbound Rules
Outbound Rules
Connection Security Rules
Monitoring
New Inbound Rule Wizard
Profile
Specify the profiles for which this rule applies.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
When does this rule apply?
Domain
Applies when a computer is connected to its corporate domain.
Private
Applies when a computer is connected to a private network location, such as a home or work place.
Public
Applies when a computer is connected to a public network location.
< Back
Next >
Cancel
Actions
Inbound Rules
New Rule...
Filter by Pr...
Filter by St...
Filter by G...
View
Refresh
Export List...
Help

6.3



6.4 IF an attacker gains access to SSH successfully they could modify or delete files, important data or even worse corrupt applications to disrupt the server's functionality.

They could also install malware in the system to maintain access to the system.

Attackers could also capture network traffic if they have access and attempt attacks like MITM in the server.

6.5 I would enforce strong password policies with complexity requirements like making use of special characters, numbers and length of words.

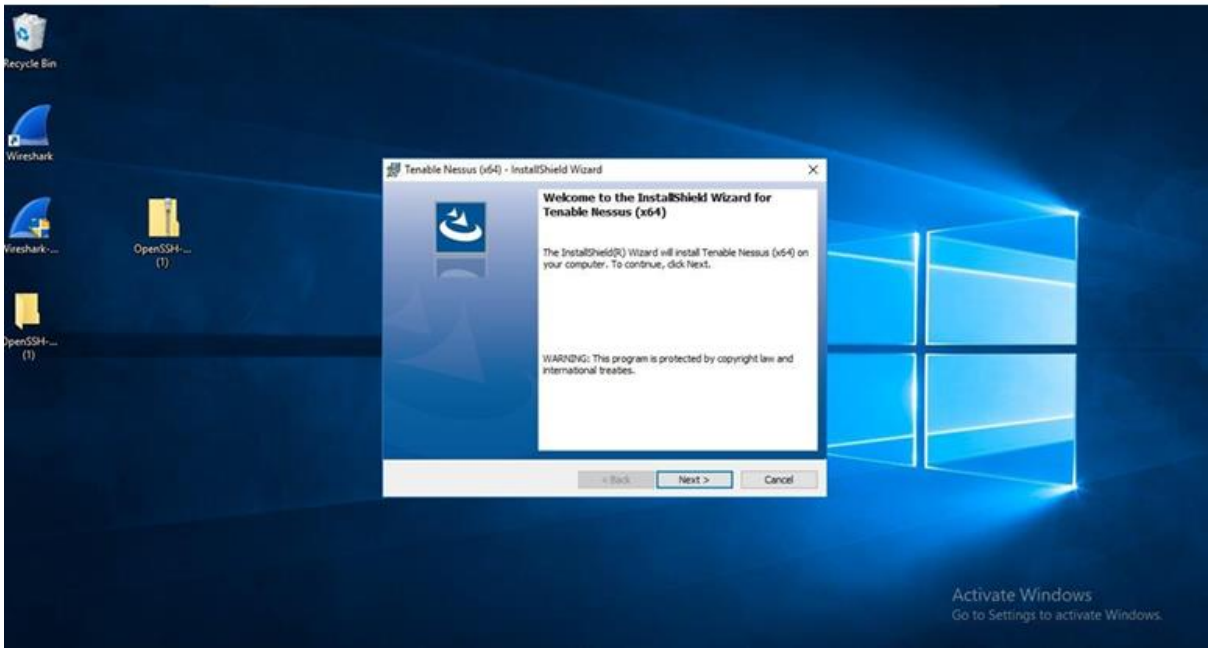
I would also disable password authentication for SSH and use public/private key based authentication instead, and limit which users can log via SSH by configuration.

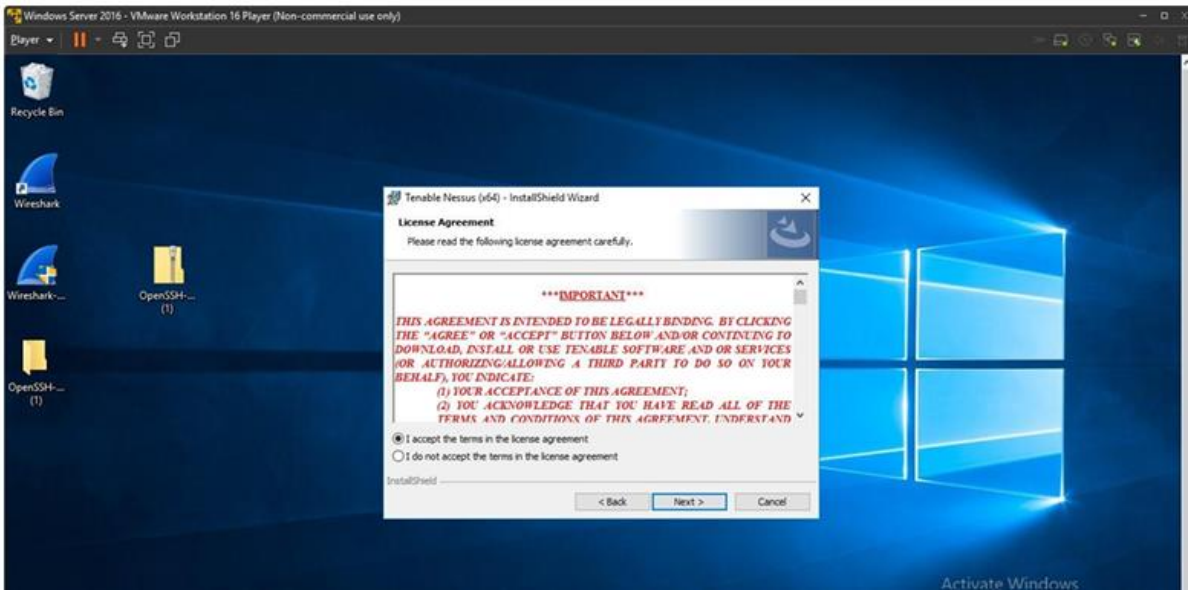
To defend against this attack, I would also restrict SSH access to trusted IP addresses via firewall rules and segment the network to limit lateral movement.

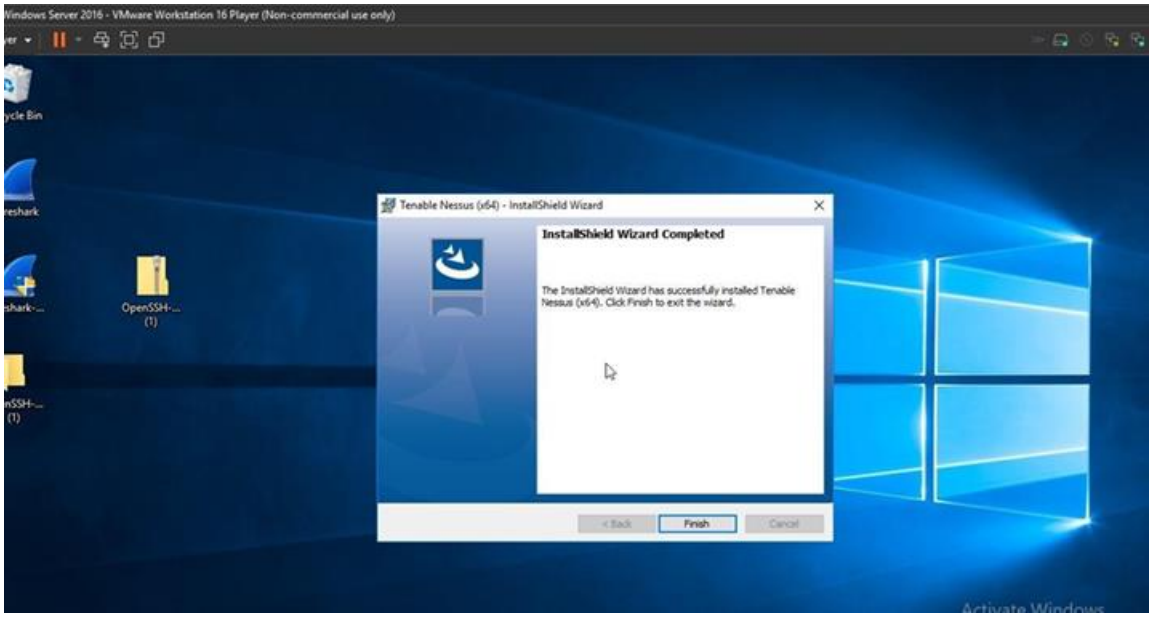
SECTION E TASK 7:

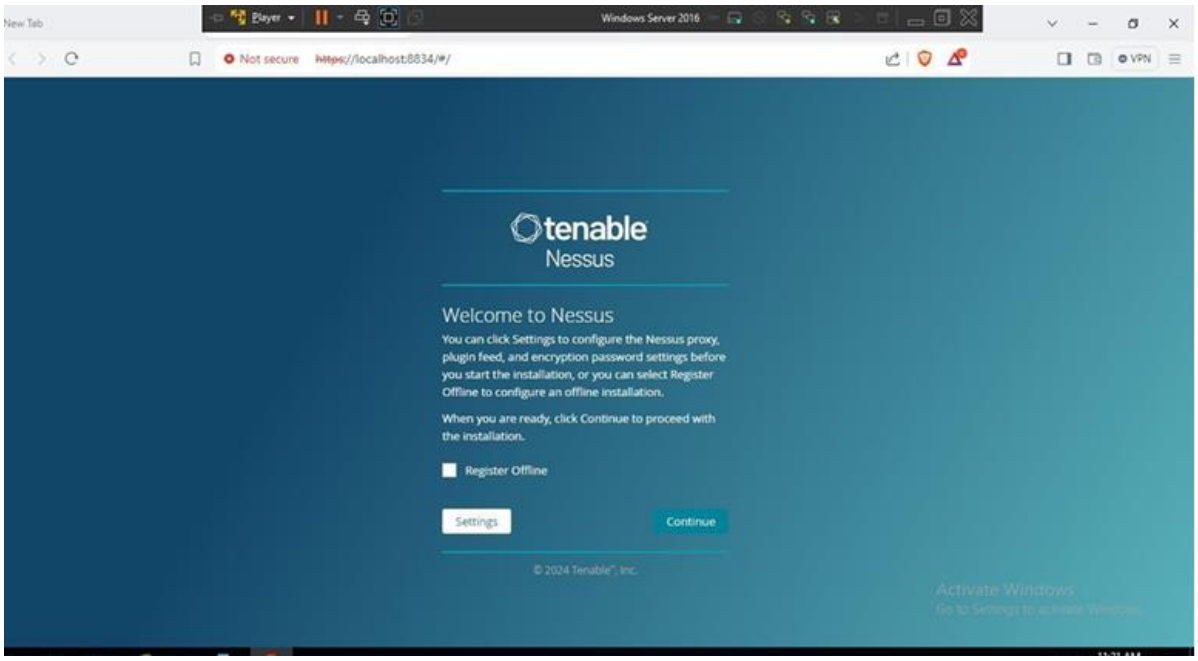
VULNERABILITY SCANNING

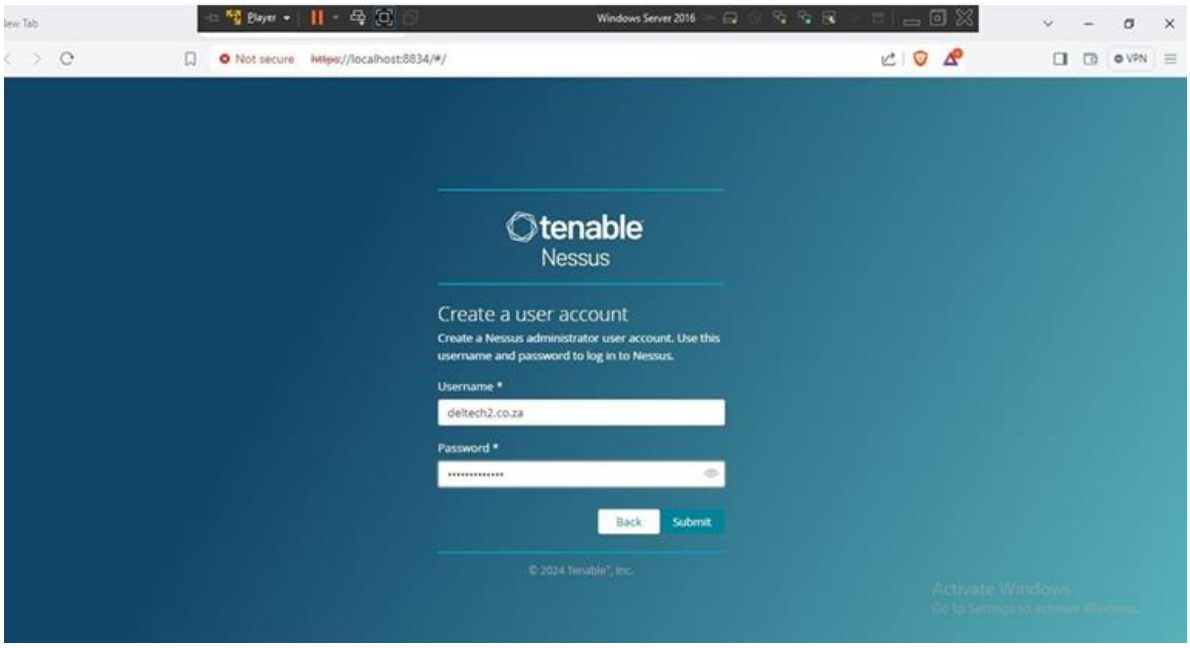
7.1

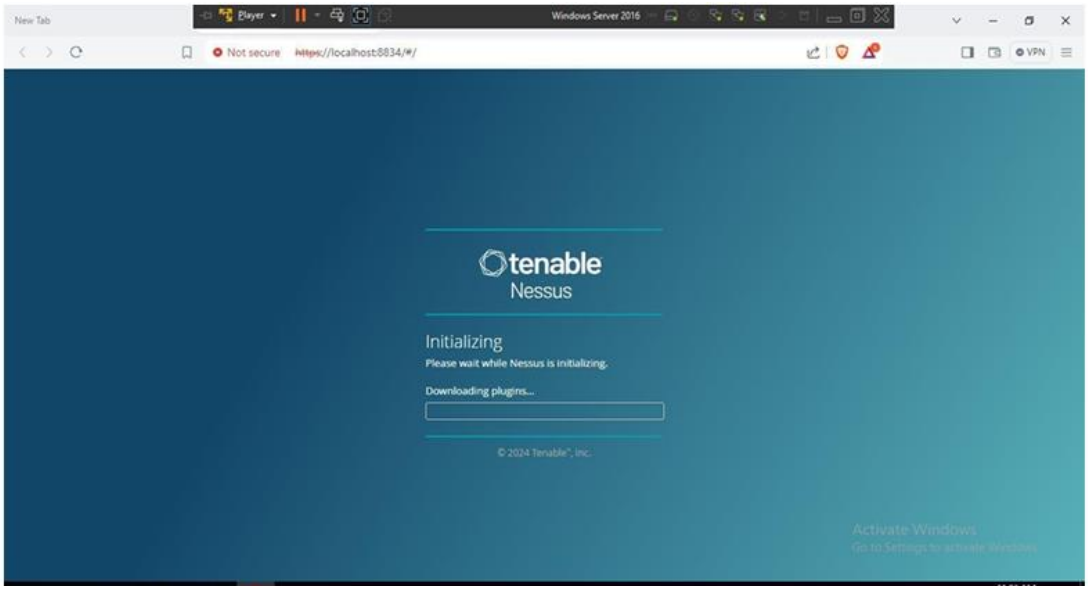





Windows Server 2016 - VMware Workstation 16 Player (Non-commercial use only)
Tenable Nessus (x64) - InstallShield Wizard
InstallShield Wizard Completed
The InstallShield Wizard has successfully installed Tenable Nessus (x64). Click Finish to exit the wizard.
< Back
Finish
Cancel


Windows Server 2016
Not secure
https://localhost:8834/#/
tenable
Nessus
Welcome to Nessus
You can click Settings to configure the Nessus proxy, plugin feed, and encryption password settings before you start the installation, or you can select Register Offline to configure an offline installation.
When you are ready, click Continue to proceed with the installation.
Register Offline
Settings
Continue
© 2024 Tenable®, Inc.



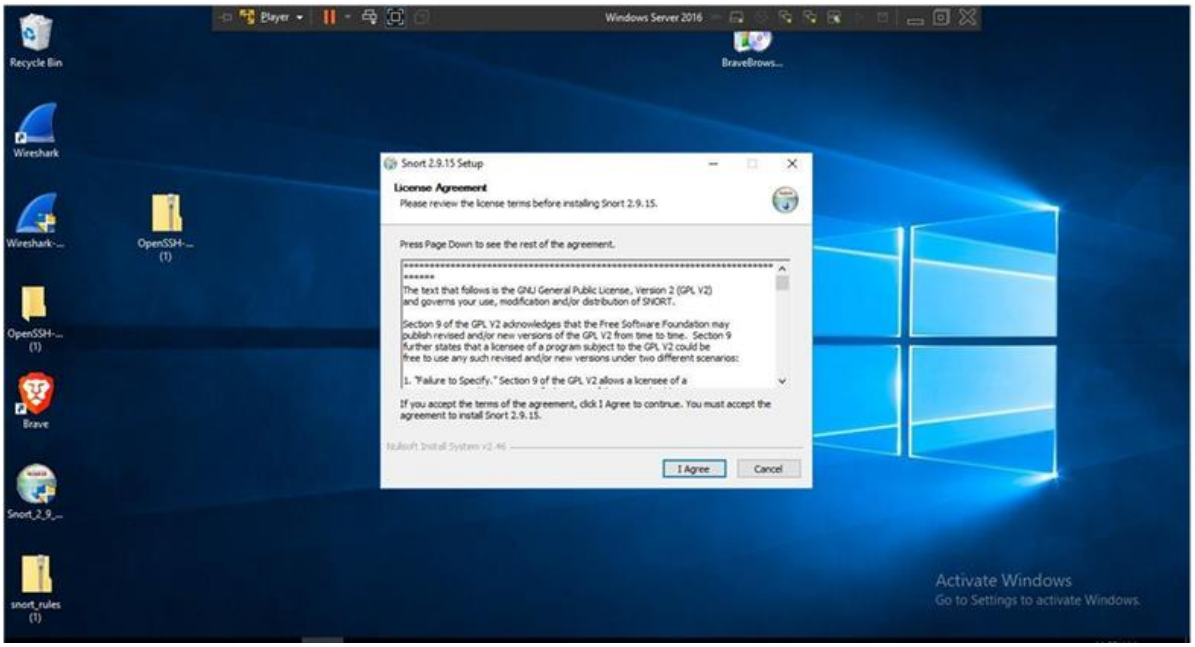


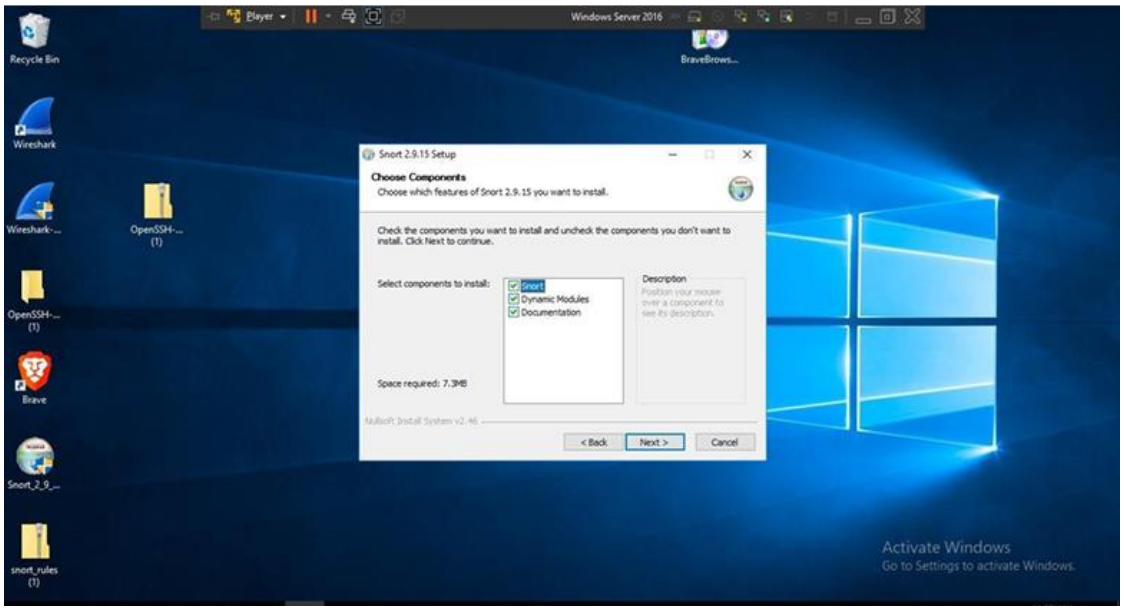
7.5 A vulnerability scan like this can be useful to a penetration test by providing a list of potential vulnerabilities in the system without actively exploiting them and it helps the testers to identify weaknesses that need further investigation.

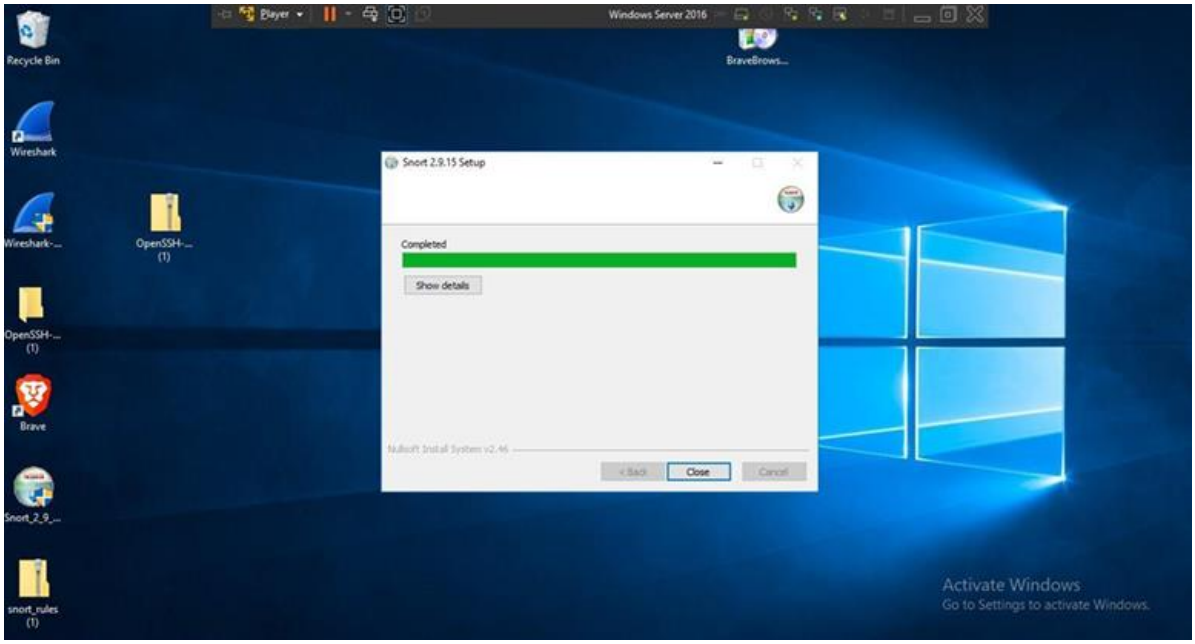
The scan provides a blueprint of the systems attack surface inkling the open ports and exposed vulnerabilities.

SECTION F TASK 8:

INTRUSION DETECTION SYSTEM






Player
Windows Server 2016
Recycle Bin
BraveBrows...
Wireshark
Wireshark-...
OpenSSH-...
(1)
OpenSSH-...
(1)
Brave
Snort_2_9_...
snort_rules
(1)
Snort 2.9.15 Setup
Completed
Show details
Nullsoft Install System v2.46
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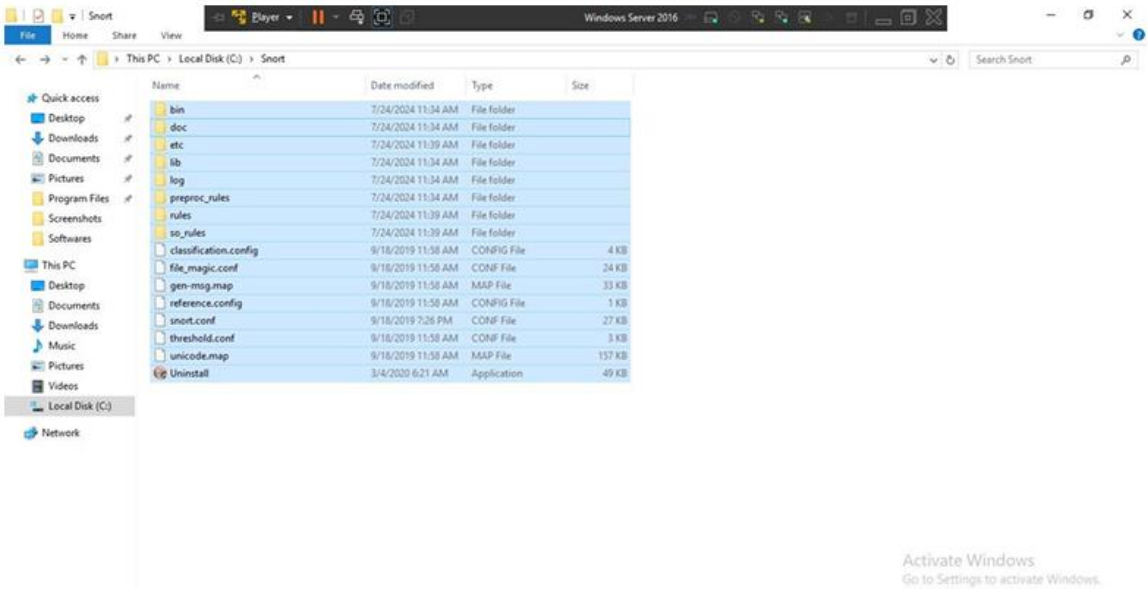
Snort
Player
Windows Server 2016
File
Home
Share
View
This PC › Local Disk (C:) › Snort
Search Snort
Quick access
Desktop
Downloads
Documents
Pictures
Program Files
Screenshots
Softwares
This PC
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
Network
Name
Date modified
Type
Size
bin 7/24/2024 11:34 AM File folder
doc 7/24/2024 11:34 AM File folder
etc 7/24/2024 11:39 AM File folder
lib 7/24/2024 11:34 AM File folder
log 7/24/2024 11:34 AM File folder
preproc_rules 7/24/2024 11:34 AM File folder
rules 7/24/2024 11:39 AM File folder
so_rules 7/24/2024 11:39 AM File folder
classification.config 9/18/2019 11:58 AM CONFIG File 4 KB
file_magic.conf 9/18/2019 11:58 AM CONF File 24 KB
gen-msg.map 9/18/2019 11:58 AM MAP File 33 KB
reference.config 9/18/2019 11:58 AM CONFIG File 1 KB
snort.conf 9/18/2019 7:26 PM CONF File 27 KB
threshold.conf 9/18/2019 11:58 AM CONF File 3 KB
unicode.map 9/18/2019 11:58 AM MAP File 157 KB
Uninstall 3/4/2020 6:21 AM Application 49 KB
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snort.conf - Notepad

File Edit Format View Help

```
#     exit with a FATAL error
#-------------------------------------------------

###################################################
# This file contains a sample snort configuration.
# You should take the following steps to create your own custom configuration:
#
#  1) Set the network variables.
#  2) Configure the decoder
#  3) Configure the base detection engine
#  4) Configure dynamic loaded libraries
#  5) Configure preprocessors
#  6) Configure output plugins
#  7) Customize your rule set
#  8) Customize preprocessor and decoder rule set
#  9) Customize shared object rule set
###################################################

###################################################
# Step #1: Set the network variables.  For more information, see README.variables
###################################################

# Setup the network addresses you are protecting
ipvar HOME_NET 192.168.10.0/24

# Set up the external network addresses. Leave as "any" in most situations
ipvar EXTERNAL_NET !$HOME_NET

# List of DNS servers on your network
ipvar DNS_SERVERS $HOME_NET

# List of SMTP servers on your network
ipvar SMTP_SERVERS $HOME_NET

# List of web servers on your network
ipvar HTTP_SERVERS $HOME_NET
```

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local - Notepad

File Edit Format View Help

```
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# Sourcefire and other third parties (the "GPL Rules") that are distributed under the
# GNU General Public License (GPL), v2.
#
# The VRT Certified Rules are owned by Sourcefire, Inc. The GPL Rules were created
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# list of third party owners and their respective copyrights.
#
# In order to determine what rules are VRT Certified Rules or GPL Rules, please refer
# to the VRT Certified Rules License Agreement (v2.0).
#
#-------------
# LOCAL RULES
#-------------
#
```

Activate Windows

local - Notepad

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```
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#
# In order to determine what rules are VRT Certified Rules or GPL Rules, please refer
# to the VRT Certified Rules License Agreement (v2.0).
#
#-------------
# LOCAL RULES
#-------------
#
# SSH Connectivity Alert
alert tcp any any -> 192.168.10.1 22 (msg:"SSH Connectivity Detected"; sid:1000001; rev:1;)
#
#TCP SYN Scan Alert
#
alert tcp any any -> 192.168.10.200 any (flags:S; msg:"SYN Scan Detected"; sid:1000002; rev:1;)
#
#Limit Output
#
alert tcp any any -> 192.168.10.1 22 (msg:"SSH Connectivity Detected"; sid:1000001; rev:1; threshold:type limit, track by_src, counnt 3, seconds
```

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